

AOS-CX 10.09 Monitoring Guide

8320, 8325, 8360 Switch Series



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This document describes features of the AOS-CX network operating system. It is intended for administrators responsible for installing, configuring, and managing Aruba switches on a network.

Applicable products

This document applies to the following products:

- Aruba 8320 Switch Series (JL479A, JL579A, JL581A)
- Aruba 8325 Switch Series (JL624A, JL625A, JL626A, JL627A)
- Aruba 8360 Switch Series (JL700A, JL701A, JL702A, JL703A, JL704C, JL705C, JL706A, JL707A, JL708A, JL709A, JL710A, JL711A, JL717C, JL718C)

Latest version available online

Updates to this document can occur after initial publication. For the latest versions of product documentation, see the links provided in [Support and Other Resources](#).

Command syntax notation conventions

Convention	Usage
<code>example-text</code>	Identifies commands and their options and operands, code examples, filenames, pathnames, and output displayed in a command window. Items that appear like the example text in the previous column are to be entered exactly as shown and are required unless enclosed in brackets ([]).
example-text	In code and screen examples, indicates text entered by a user.
Any of the following: ■ <code><example-text></code> ■ <code><example-text></code> ■ <i>example-text</i> ■ <i>example-text</i>	Identifies a placeholder—such as a parameter or a variable—that you must substitute with an actual value in a command or in code: ■ For output formats where italic text cannot be displayed, variables are enclosed in angle brackets (< >). Substitute the text—including the enclosing angle brackets—with an actual value. ■ For output formats where italic text can be displayed, variables might or might not be enclosed in angle brackets. Substitute the text including the enclosing angle brackets, if any, with an actual value.
	Vertical bar. A logical OR that separates multiple items from which you can choose only one. Any spaces that are on either side of the vertical bar are included for readability and are not a required part of the command syntax.

Convention	Usage
{ }	Braces. Indicates that at least one of the enclosed items is required.
[]	Brackets. Indicates that the enclosed item or items are optional.
... or ...	Ellipsis: ■ In code and screen examples, a vertical or horizontal ellipsis indicates an omission of information. ■ In syntax using brackets and braces, an ellipsis indicates items that can be repeated. When an item followed by ellipses is enclosed in brackets, zero or more items can be specified.

About the examples

Examples in this document are representative and might not match your particular switch or environment.

The slot and port numbers in this document are for illustration only and might be unavailable on your switch.

Understanding the CLI prompts

When illustrating the prompts in the command line interface (CLI), this document uses the generic term `switch`, instead of the host name of the switch. For example:

```
switch>
```

The CLI prompt indicates the current command context. For example:

```
switch>
```

Indicates the operator command context.

```
switch#
```

Indicates the manager command context.

```
switch(CONTEXT-NAME)#
```

Indicates the configuration context for a feature. For example:

```
switch(config-if)#
```

Identifies the interface context.

Variable information in CLI prompts

In certain configuration contexts, the prompt may include variable information. For example, when in the VLAN configuration context, a VLAN number appears in the prompt:

```
switch(config-vlan-100)#
```

When referring to this context, this document uses the syntax:

```
switch(config-vlan-<VLAN-ID>)#
```

Where `<VLAN-ID>` is a variable representing the VLAN number.

Identifying switch ports and interfaces

Physical ports on the switch and their corresponding logical software interfaces are identified using the format:

```
member/slot/port
```

On the 83xx Switch Series

- *member*: Always 1. VSF is not supported on this switch.
- *slot*: Always 1. This is not a modular switch, so there are no slots.
- *port*: Physical number of a port on the switch.

For example, the logical interface 1/1/4 in software is associated with physical port 4 on the switch.



If using breakout cables, the port designation changes to x:y, where x is the physical port and y is the lane when split to 4 x 10G or 4 x 25G. For example, the logical interface 1/1/4:2 in software is associated with lane 2 on physical port 4 in slot 1 on member 1.

Identifying modular switch components

- Power supplies are on the front of the switch behind the bezel above the management modules. Power supplies are labeled in software in the format: *member/power supply*:
 - *member*: 1.
 - *power supply*: 1 to 4.
- Fans are on the rear of the switch and are labeled in software as: *member/tray/fan*:
 - *member*: 1.
 - *tray*: 1 to 4.
 - *fan*: 1 to 4.
- Fabric modules are not labeled on the switch but are labeled in software in the format: *member/module*:
 - *member*: 1.
 - *member*: 1 or 2.
- The display module on the rear of the switch is not labeled with a member or slot number.

Diagnosing with the LEDs

This section describes LED patterns on the switch that indicate problem conditions for general switch operation troubleshooting.

1. Check the table for the LED pattern you see on the switch.
2. Refer to the corresponding diagnostic tip.

Table 1: LED error indicators for 8320

Global status	Port LED	Diagnostic tip
Off with power cord plugged in	N/A	1
Solid amber	N/A	2
Slow flash amber	N/A	3
Slow flash amber	Slow flash amber*	4
Solid green	Off with cable connected	5
Solid green	On, but the port is not communicating	6

*The flashing behavior is an on/off cycle approximately once every 1.6 seconds.

Table 2: LED error indicators for 8325

PS1/PS2 LEDs	Global Status	Fan	Port LED	Diagnostic Tip
Off with power cords plugged in	-	-	-	1
On amber**	Flashing amber	-	-	2
On green	Flashing amber	On amber	-	3
On green	Flashing amber	-	Flashing amber	4
On green	On green	-	Off with cable connected	5
On green	On green	-	On, but the port is not communicating	6

**Either the PS1 or PS2 LED is on amber, but not both.

Table 3: Diagnostic tips

Tip	Problem	Solution
1	Both switch power supplies are not plugged into an active AC power source.	Verify the AC power source works by plugging another device into the outlet. Or try plugging the power supplies into different outlets or try different power cords. If the problem is still not resolved, both power supplies may be faulty.
2	One of the power supplies is not plugged into an active A power source, or the power supply may have failed.	Verify that the power cord is plugged into an active power source and to the power supply. Make sure that the connections are snug. Try power cycling the switch by unplugging and plugging the power cord back into the other working power supply. If the PS1/PS2 LED is still not on, verify the AC power source works by plugging another device into the outlet or try a different power cord. If the power source and power cord are OK and this condition persists, the switch power supply may have failed. Call your Hewlett Packard Enterprise-authorized network reseller, or use the electronic support services from Hewlett Packard Enterprise to get assistance.
3	One of the switch fan assemblies may have failed.	Try disconnecting power from the switch and wait a few moments. Then reconnect the power to the switch and check the LEDs again. If the error indication reoccurs, one of the fan assemblies has failed. If the ambient temperature does not exceed normal room temperature, the switch may continue to operate under this condition; but for best operation, replace the fan assembly. Call your Hewlett Packard Enterprise-authorized network reseller, or use the electronic support services from Hewlett Packard Enterprise to get assistance.
4	The network port for which the LED is flashing has experienced a self-test or initialization failure.	Try power cycling the switch. If the fault indication reoccurs: ■ There may be a port configuration mismatch where a 10G transceiver is installed in a port configured for 25G, or the reverse. ■ A 10GBase-T transceiver may be installed in an incompatible port. Only ports 1, 2, 4, 5, 7, 8, 10, and 11 support 10GBase-T transceivers. ■ The transceiver may have failed. ■ The switch port may have failed. Check the switch Event Log and show interface

Tip	Problem	Solution
		command output for indication of the fault condition. If the port is an SFP+/SFP28 transceiver or QSFP+/QSFP28 transceiver, verify that it is one of the transceivers supported by the switch. Unsupported or unrecognized transceivers will be identified with this fault condition. For a list of supported transceivers, see the <i>Transceiver Guide</i> in the Aruba Support Portal. The transceivers are also tested when they are "hot-swapped" - installed or changed while the switch is powered on. To verify that the port has failed, remove and reinstall the transceiver without powering off the switch. If the port fault indication reoccurs, you will have to replace the transceiver. Check the event log to see why the transceiver failed. To get assistance, call your Hewlett Packard Enterprise-authorized network reseller, or use the electronic support services from Hewlett Packard Enterprise.
5	The network connection is not working properly.	Try the following procedures: ■ For the indicated port, verify that both ends of the cabling, at the switch and the connected device, are connected properly. ■ Verify that the connected device and switch are both powered on and operating correctly. ■ Verify that you have used the correct cable type for the connection: ◦ For fiber-optic connections, verify that the transmit port on the switch is connected to the receive port on the connected device and that the switch receive port is connected to the transmit port on the connected device. ◦ The cable verification process must include all patch cables from any end devices, including the switch, to any patch panels in the cabling path. ■ Verify that the port has not been disabled through a switch configuration change. Use the console interface or, if you have configured an IP address on the switch, use the web browser interface to determine the state of the port and re-enable the port if necessary. ■ Verify that the switch port configuration matches the configuration of the attached device. For example, if the switch port is configured as "Full-

Tip	Problem	Solution
		duplex”, the port on the attached device also MUST be configured as “Full-duplex”. If the configurations do not match, the results could be an unreliable connection, or no link at all. ■ If the other procedures do not resolve the problem, try using a different port or a different cable.
6	The port may be improperly configured, or the port may be in a “blocking” state by the normal operation of the Spanning Tree, LACP, or IGMP features.	Use the switch console to see if the port is part of a dynamic trunk (through the LACP feature), if Spanning Tree is enabled on the switch, and if the port may have been put into a “blocking” state by those features. The <code>show lacp interfaces</code> command displays the port status for the LACP feature; the <code>show spanning tree</code> command displays the port status for Spanning Tree. Also check the Port Status screen using the <code>show interfaces</code> command to see if the port has been configured as “disabled”. Other switch features that may affect the port operation include VLANs, IGMP, and port group settings. Use the switch console to see how the port is configured for these features. Ensure that the device at the other end of the connection is indicating a good link to the switch. If it is not, the problem may be with the cabling between the devices or the connectors on the cable.

Chassis LEDs

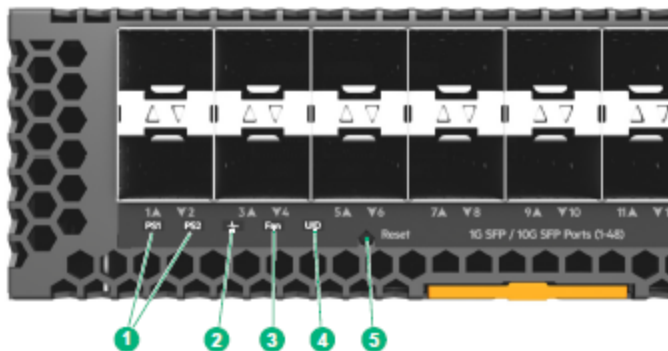


Table 1: Chassis LED labels

	LED
1	Power supply LEDs
2	Global status LEDs
3	Fan LED
4	Unit identification LED
5	Reset button

Table 2: Chassis LED behavior

Chassis LEDs	Function	State	Meaning
PS1/PS2	Power supply status	On green	Power supply is installed and operating normally.
		Slow flash amber	Fault detected for installed power supply.
		Off	Power supply is not installed or not receiving power.
Fan	Fan tray status	On green	System fans are operating normally.
		Slow flash amber	One or more system fans have a fault or the minimum number of fans are not installed.

Chassis LEDs	Function	State	Meaning
Global Status	Internal power status of the switch. Self-test status Switch/port fault status	On amber	The switch has passed self-test and is powered up normally.
		Slow flash green*	The switch self-test and initialization are in progress after the switch has been power cycled or reset. The switch is not operational until this LED stops blinking green.
		Slow flash amber**	A fault or initialization failure has occurred on the switch, one of the switch ports, OOBM port, USB port, console port, power supplies, or a fan. The Status LED for the component with the fault will flash simultaneously.
		Off	The unit is not receiving power.
UID (Unit Identification)	Used to identify a unit in a rack or collection of products.	On or slow flash^	The <code>LED locator</code> command allows you to flash or turn on the LED. The default is 30 minutes.
		Off	LED will clear after the timeout period has expired.

*The slow flash behavior is an on/off cycle approximately every 1.6 seconds.

**The slow flash behavior is an on/off cycle approximately every 1.6 seconds.

^The slow flash behavior is an on/off cycle approximately every 1.6 seconds.

Port LEDs

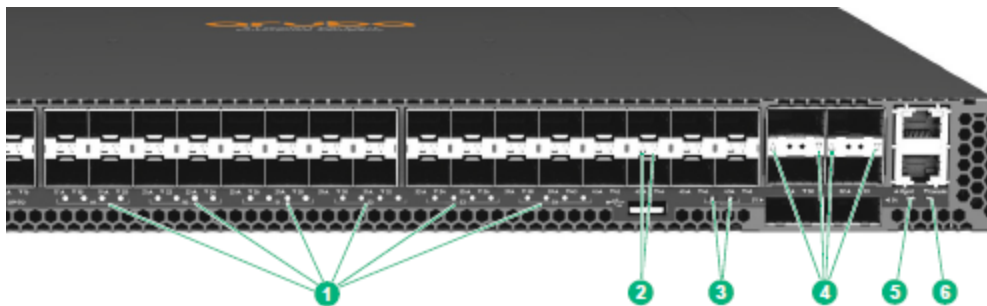


Table 1: Port LED labels

	LED
1	QSFP+ port lane LEDs — these LEDs are not used by the product and should remain off throughout the product's operation

	LED
2	SFP+ port LEDs
3	QSFP+ port 51, 54 LEDs
4	QSFP+ port 49, 50, 52, 53 LEDs
5	Out-of-band management port Link LED
6	Out-of-band management port Act (activity) LED

Table 2: *Port LED behavior*

Chassis LEDs	Function	State	Meaning
SFP+ port LEDs	To display link and activity information for the port.	On/flashing green	Shows a valid link at 1 Gbps or 10 Gbps. Flashing indicates port activity.
		Slow flash amber	When the Global Status LED is flashing amber, indicates an unsupported transceiver or a port failure.
QSFP+ port LEDs	To display link and activity information for the port.	On/flashing green	Shows a valid link at 40 Gbps. Flashing indicates port activity.
		Off	When the Global Status LED is flashing amber, indicates an unsupported transceiver or a port failure.
Management port Link LED	To display link information for the port.	On green	Shows a valid link.
Management port Act LED	To display activity information for the port.	Flashing green	Flashing indicates port activity.

Chassis LEDs

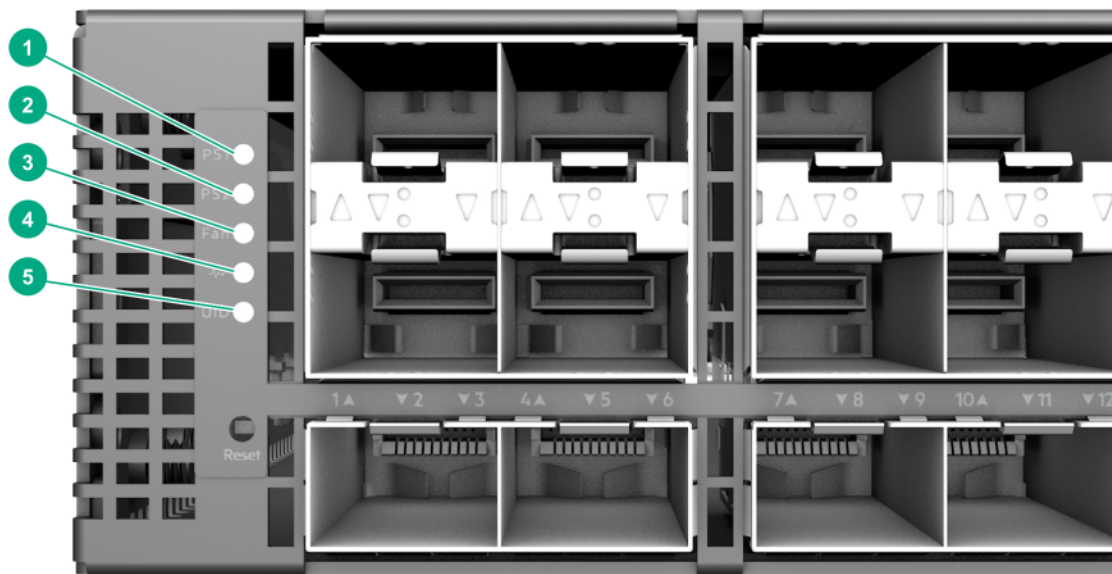


Table 1: Chassis LEDs for the Aruba 8325-48Y8C (JL624A and JL625A)

	LED
1	Power supply 1 (PS1) LEDs
2	Power supply 2 (PS2) LEDs
3	Fan LED
4	Global status LED
5	Unit identification LED

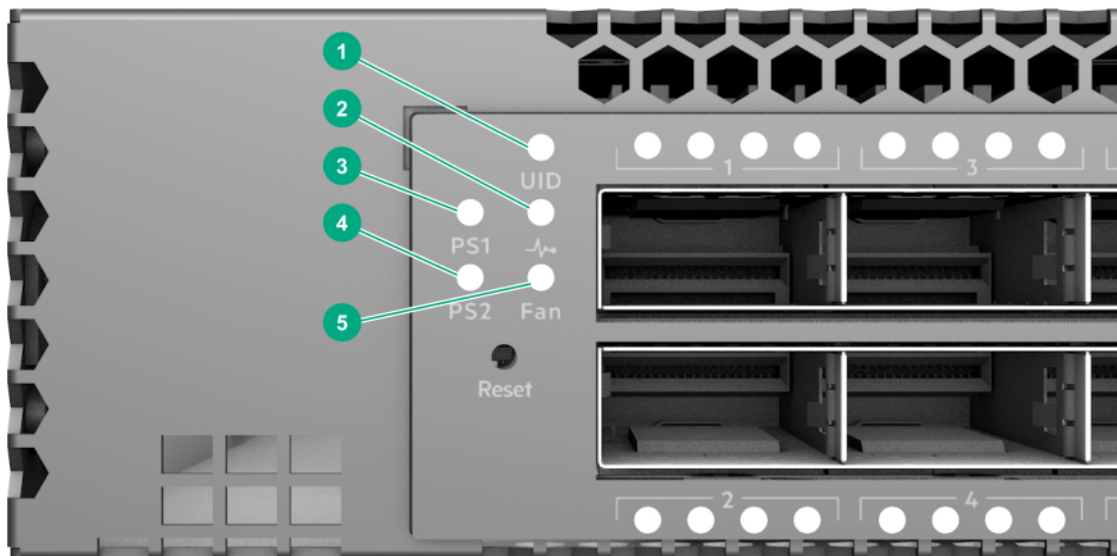


Table 2: Chassis LEDs for the Aruba 8325-32C (JL626A and JL627A)

	LED
1	Unit identification LED
2	Global status LED
3	Power supply 1 (PS1) LEDs
4	Power supply 2 (PS2) LEDs
5	Fan LED

Table 3: Chassis LED behavior

Chassis LEDs	Function	State	Meaning
PS1/PS2	Power supply status	On green	Power supply is installed and operating normally.
		On amber	Fault detected for installed power supply, or power supply is not receiving power.
		Off	Power supply is not installed.
Fan	Fan tray status	On green	System fans are operating normally.
		On amber	One or more system fans have a fault or the minimum number of fans are not installed.
Global Status	Internal power status of the switch. Self-test status Switch/port fault status	On green	The switch has passed self-test and is powered up normally.
		Flashing amber	The switch initialization is in progress during boot up.

Chassis LEDs	Function	State	Meaning
			- A fault or initialization failure has occurred on the switch, one of the switch ports, power supplies, or a fan. - The port LEDs with the fault will flash simultaneously. LEDs for power supplies and fans with a fault will be on amber. - Port-speed mismatch. A transceiver is installed in a port configured for a different speed.
		Off	The unit is not receiving power.
UID (Unit Identification)	Used to identify a unit in a rack or collection of products.	On blue or flashing blue	The <code>LED locator on</code> command allows you turn on the LED. The default is 30 minutes. The <code>LED locator flashing</code> command will flash the LED.
		Off	The <code>LED locator off</code> command turns off the LED.

Port LEDs

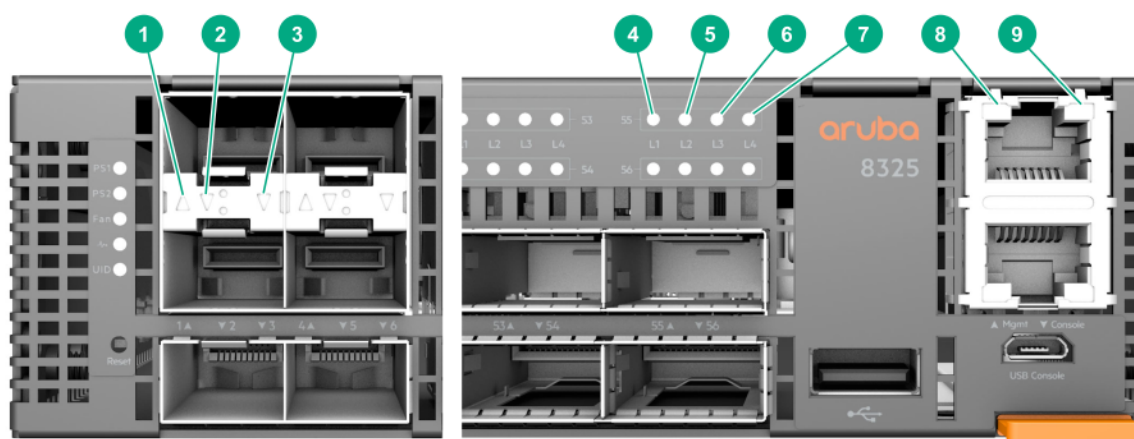


Table 1: Port LEDs for the Aruba 8325-48Y8C (JL624A and JL625A)

	LED
1	Upper SFP28 port LED
2	Middle SFP28 port LED
3	Lower SFP28 port LED
4	QSFP28 port LED and lane 1 indicator

	LED
5	QSFP28 lane 2 LED
6	QSFP28 lane 3 LED
7	QSFP28 lane 4 LED
8	Out-of-band management port Link LED
9	Out-of-band management port Act (activity) LED

Table 2: Port LED behavior for the Aruba 8325-48Y8C (JL624A and JL625A)

Chassis LEDs	Function	State	Meaning
SFP28 port LEDs	To display link and activity information for the port.	On/flashing green	Shows a valid link at 25/10 Gbps. - Fast flashing* indicates port activity at 25 Gbps. - Slow flashing** indicates port activity at 10 Gbps.
		Flashing amber	When the Global Status LED is simultaneously flashing amber, indicates port-speed mismatch, an incompatible, unsupported, faulty transceiver, or a port failure.
		Off	Port is disabled, not connected, or not receiving a link beat signal.
QSFP28 port LEDs	To display link and activity information for the port.	On/flashing green	Shows a valid link at 100/40 Gbps. - Fast flashing^ indicates port activity at 100 Gbps. - Slow flashing^^ indicates port activity at 40 Gbps.
		Flashing amber	When the Global Status LED is simultaneously flashing amber with the Lane 1 LED, indicates an unsupported or faulty transceiver, or a port failure.
		Off	Port is disabled, not connected, or not receiving a link beat signal. Lanes 2-4 are always off and are currently unused by HPE-Aruba software.

Chassis LEDs	Function	State	Meaning
Management port Link LED	To display link information for the port.	On green	Shows a valid link.
		Off	Port is disabled, not connected, or not receiving a link beat signal.
Management port Act LED	To display activity information for the port.	Flashing yellow	Flashing indicates port activity.

*The fast flashing behavior is an on/off cycle once every 0.8 seconds, approximately.

**The slow flashing behavior is an on/off cycle once every 1.6 seconds, approximately.

^The fast flashing behavior is an on/off cycle once every 0.8 seconds, approximately.

^^The slow flashing behavior is an on/off cycle once every 1.6 seconds, approximately.

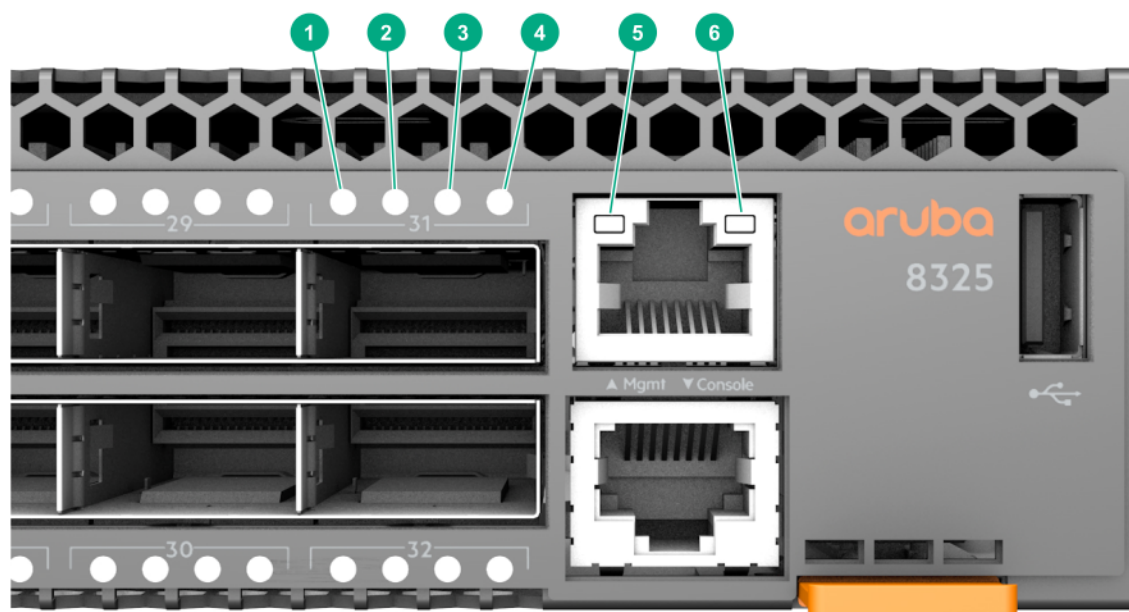


Table 3: Port LEDs for the Aruba 8325-32C (JL626A and JL627A)

	LED
1	QSFP28 port LED and lane 1 indicator
2	QSFP28 lane 2 LED (Not supported with currently released software.)
3	QSFP28 lane 3 LED (Not supported with currently released software.)
4	QSFP28 lane 4 LED (Not supported with currently released software.)
5	Unused
6	Out-of-band management port Link and Activity LED

Table 4: Port LED behavior for the Aruba 8325-32C (JL626A and JL627A)

Chassis LEDs	Function	State	Meaning
QSFP28 port LEDs	To display link and activity information for the port.	On/flashing green	Shows a valid link at 100/40 Gbps. - Fast flashing* indicates port activity at 100 Gbps. - Slow flashing** indicates port activity at 40 Gbps.
		Flashing amber	When the Global Status LED is simultaneously flashing amber with the Lane 1 LED, indicates an unsupported or faulty transceiver, or a port failure.
		Off	Port is disabled, not connected, or not receiving a link beat signal. Lanes 2-4 are always off and are currently unused by HPE-Aruba software.
Management port Link LED	To display link information for the port.	On green	Shows a valid link.
		Off	Port is disabled, not connected, or not receiving a link beat signal.
Management port Act LED	To display activity information for the port.	Flashing yellow	Flashing indicates port activity.

*The fast flashing behavior is an on/off cycle once every 0.8 seconds, approximately.

**The slow flashing behavior is an on/off cycle once every 1.6 seconds, approximately.

Alarm commands

alarm

```
alarm {temperature | power supply} [action <LOG-AND-TRAP> | <RELAY>]
no alarm {temperature | power supply} [action <LOG-AND-TRAP> | <RELAY>]
```

Description

Configures global events to be forwarded to the output alarm port. The `no` form of this command disables the specified configuration.

Parameter	Description
<code>{temperature}</code>	Selects the alarm for ambient temperatures reaching threshold limits. The threshold is 70°C.
<code>{power supply}</code>	Selects the alarm events from the power supply.
<code><LOG-AND-TRAP></code>	Generates an event log and SNMP trap.
<code><RELAY></code>	Relays an event to alarm output port.

Examples

Disabling a temperature event to remove the configuration for all actions associated with the event:

```
switch(config)# no alarm temperature
```

Configuring an alarm for a temperature event for the log-and-trap action:

```
switch(config)# alarm temperature action log-and-trap
```

Removing the configuration for the temperature event for a log-and-trap action:

```
switch(config)# no alarm temperature action log-and-trap
```

Configuring an alarm for a power-supply event for the relay action:

```
switch(config)# alarm power-supply action relay
```

Removing the configuration for the power-supply event for the relay action:

```
switch(config)# no alarm power-supply action relay
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

alarm input

```
alarm input {IN1 | IN2} [name <STRING>] [action <LOG-AND-TRAP> | <RELAY>] [trigger  
<CLOSED | OPEN>]  
no alarm input {IN1 | IN2} [name <STRING>] [action <LOG-AND-TRAP> | <RELAY>] [trigger  
<CLOSED | OPEN>]
```

Description

Configures input alarm and actions. The `no` form of this command removes the specified configuration.

Parameter	Description
{IN1 IN2}	Specifies the input alarm port..

Parameter	Description
name	Specifies the external device connected.
<STRING>	Descriptive string.
action	Specifies the action to be taken when the monitored event occurs.
<LOG-AND-TRAP>	Generates an event log and SNMP trap.
<RELAY>	Relays an event to alarm output port.
trigger	Triggers an alarm based on a normally open or closed circuit.
<CLOSED>	Generates an alarm event when the circuit is closed.
<OPEN>	Generates an alarm event when the circuit is open.

Examples

Configuring an alarm on input port 1 named Door-Sensor:

```
switch(config)# alarm input IN1 name Door-Sensor
```

Removing the configuring for an alarm on input port 1:

```
switch(config)# no alarm input IN1
```

Configuring an alarm on input port 1 with log-and-trap action:

```
switch(config)# alarm input IN1 action log-and-trap
```

Removing the configuration for an alarm on input port 1 with log-and-trap action:

```
switch(config)# no alarm input IN1 action log-and-trap
```

Configuring an alarm on input port 1 to trigger an alarm when the door sensor circuit is closed:

```
switch(config)# alarm input IN1 trigger closed
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

alarm snooze

alarm snooze [time in minutes] [repeat]

Description

Configures any active event forwarded to alarm relay to be snoozed.

Parameter	Description
time in minutes	Specifies the value for time in minutes. The range is 0-1440 minutes..
repeat	Repeats the previous snooze time.

Examples

Configuring an alarm relay action for 10 minutes:

```
switch(config)# alarm snooze 10
```

Configuring an alarm relay action to be repeated with previously configured snooze time:

```
switch(config)# alarm snooze repeat
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	config	Administrators or local user group members with execution rights for this command.

show alarm

show alarm [temperature | power supply]

Description

Shows all of the details of global status monitoring alarm events.

Parameter	Description
temperature	Selects the alarm for ambient temperatures reaching threshold limits.
power supply	Selects the alarm events from the power supply.

Examples

Showing details for all global alarm events on the switch:

```
switch# show alarm
Alarm Snooze Timer Status: active
Duration remaining: 1 min 32 sec
-----
Global Alarm: Temperature
-----
Alarm Event      Status      log-and-trap  Relay
-----
Temperature      inactive    true          false

-----
Global Alarm: power-supply
-----
Alarm Event      Status      log-and-trap  Relay
-----
power-supply     active      false         true
```

Showing details for the temperature alarm:

```
switch# show alarm temperature
Alarm Snooze Timer Status: Active
Duration remaining: 5 min 36 sec
-----
Global Alarm: Temperature
-----
Alarm Event      Status      log-and-trap  Relay
-----
Temperature      inactive    true          false
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

show alarm input

```
show alarm input [IN1 | IN2]
```

Description

Shows the alarm details for all ports or for the specified port.

Parameter	Description
{IN1 IN2}	Specifies the input alarm port..

Examples

Showing details for all alarm input ports on the switch:

```
switch# show alarm input
Alarm Snooze Timer Status: inactive
-----
Input Alarm IN1, Name: Door-Sensor
-----
Alarm Port      Status      log-and-trap  Relay      Trigger
-----
IN1              inactive    true          false      closed

-----
Input Alarm IN2, Name: N/A
-----
Alarm Port      Status      log-and-trap  Relay      Trigger
-----
IN2              active      false         true       open
```

Showing details for alarm input ports IN1:

```
switch# show alarm input IN1
Alarm Snooze Timer Status: inactive
-----
Input Alarm IN1, Name: Door-Sensor
-----
Alarm Port      Status      log-and-trap  Relay      Trigger
-----
IN1              inactive    true          false      closed
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

show alarm timer

```
show alarm timer
```

Description

Shows the status of an alarm snooze timer's status and duration.

Examples

Showing the status of an alarm snooze timer when it is inactive :

```
switch# show alarm timer
Alarm Snooze Timer Status: inactive
```

Showing the status of an active 3-minute alarm snooze timer:

```
switch# show alarm timer
Alarm Snooze Timer Status: active
Duration remaining: 2 min 55 sec
```

Command History

Release	Modification
10.08	Featured introduced.

Command Information

Platforms	Command context	Authority
4100i	Manager (#)	Administrators or local user group members with execution rights for this command.

boot fabric-module

boot fabric-module <SLOT-ID>

Description

Reboots the specified fabric module.

Parameter	Description
<SLOT-ID>	Specifies the member and slot of the module in the format member/slot. For example, to specify the module in member 1 slot 3, enter 1/3.

Usage

The `boot fabric-module` command reboots the specified fabric module. Traffic performance is affected while the module is down.

If the specified module is the only fabric module in an up state, rebooting that module stops traffic switching between line modules and the line modules power down. The line modules power up when one fabric module returns to an up state.

This command is valid for fabric modules only.

Examples

Rebooting the fabric module in slot **1/3** when auto-confirm is not enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n)? y

switch#
```

Rebooting the fabric module in slot **1/1** when auto-confirm is enabled:

```
switch# boot fabric-module 1/3
This command will reboot the specified fabric module. Traffic performance may
be affected while the module is down. Rebooting the last fabric module will
stop traffic switching between line modules.
Do you want to continue (y/n) y (auto-confirm)

switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
	Manager (#)	Administrators or local user group members with execution rights for this command.

boot line-module

`boot line-module <SLOT-ID>`

Description

Reboots the specified line module.

Parameter	Description
<code><SLOT-ID></code>	Specifies the member and slot of the module in the format member/slot. For example, to specify the module in member 1 slot 3, enter 1/3.

Usage

Reboots the specified line module. Any traffic for the switch passing through the affected module (SSH, TELNET, and SNMP) is interrupted. It can take up to 2 minutes to reboot the module. During that time, you can monitor progress by viewing the event log.

This command is valid for line modules only.

Examples

Reloading the module in slot 1/1:

```
switch# boot line-module 1/1
This command will reboot the specified line module and interfaces on this
module will not send or receive packets while the module is down. Any
traffic passing through the line module will be interrupted. Management
sessions connected through the line module will be affected. It might take
up to 2 minutes to complete rebooting the module. During that time, you can
monitor progress by viewing the event log.
Do you want to continue (y/n)? y
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
	Manager (#)	Administrators or local user group members with execution rights for this command.

boot management-module

`boot management-module {active | standby | <SLOT-ID>}`

Description

Reboots the specified management module. Choose the management module to reboot by role (active or standby) or by slot number.

Parameter	Description
<i>active</i>	Selects the active management module.
<i>standby</i>	Selects the standby management module.
<i><SLOT-ID></i>	Specifies the member and slot of the management module in the format <code>member/slot</code> . For example, to specify the module in member 1 slot 5, enter <code>1/5</code> .

Usage

This command reboots a single management module in a chassis. Choose the management module to reboot by role (active or standby) or by slot number.

You can use the `show images` command to show information about the primary and secondary system images.

If you reboot the active management module and the standby management module is available, the active management module reboots and the standby management module becomes the active management module.

If you reboot the active management module and the standby management module is not available, you are warned, you are prompted to save the configuration, and you are prompted to confirm the operation.

If you reboot the standby management module, the standby management module reboots and remains the standby management module.

If you attempt to reboot a management module that is not available, the `boot` command is aborted.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot` command is aborted.



Hewlett Packard Enterprise recommends that you use the `boot management-module` command instead of pressing the module reset button to reboot a management module because if you are rebooting the only available management module, the `boot management-module` command enables you to save the configuration, cancel the reboot, or both.

Examples

Rebooting the active management module when the standby management module is available:

```
switch# boot management-module active
The management-module in slot 1/5 is going down for reboot now.
```

Rebooting the active management module when the standby management module is not available:

```
switch# boot management-module 1/5
The management module in slot 1/5 is currently active and no
standby management module was found.
This will reboot the entire switch.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
	Manager (#)	Administrators or local user group members with execution rights for this command.

boot set-default

```
boot set-default {primary | secondary}
```

Description

Sets the default operating system image to use when the system is booted.

Parameter	Description
primary	Selects the primary network operating system image.
secondary	Selects the secondary network operating system image.

Example

Selecting the primary image as the default boot image:

```
switch# boot set-default primary
Default boot image set to primary.
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

boot system

```
boot system [primary | secondary | serviceos]
```

Description

Reboots all modules on the switch. By default, the configured default operating system image is used. Optional parameters enable you to specify which system image to use for the reboot operation and for future reboot operations.

Parameter	Description
<code>primary</code>	Selects the primary operating system image for this reboot and sets the configured default operating system image to <code>primary</code> for future reboots.
<code>secondary</code>	Selects the secondary operating system image for this reboot and sets the configured default operating system image to <code>secondary</code> for future reboots.
<code>serviceos</code>	Selects the service operating system for this reboot. Does not change the configured default operating system image. The service operating system acts as a standalone bootloader and recovery OS for switches running the AOS-CX operating system and is used in rare cases when troubleshooting a switch.

Usage

This command reboots the entire system. If you do not select one of the optional parameters, the system reboots from the configured default boot image.

You can use the `show images` command to show information about the primary and secondary system images.

Choosing one of the optional parameters affects the setting for the default boot image:

- If you select the `primary` or `secondary` optional parameter, that image becomes the configured default boot image for future system reboots. The command fails if the switch is not able to set the operating system image to the image you selected.

You can use the `boot set-default` command to change the configured default operating system image.

- If you select `serviceos` as the optional parameter, the configured default boot image remains the same, and the system reboots all management modules with the service operating system.

If the configuration of the switch has changed since the last reboot, when you execute the `boot system` command you are prompted to save the configuration and you are prompted to confirm the reboot operation.

Saving the configuration is not required. However, if you attempt to save the configuration and there is an error during the save operation, the `boot system` command is aborted.

Examples

Rebooting the system from the configured default operating system image:

```
switch# boot system
Do you want to save the current configuration (y/n)? y
The running configuration was saved to the startup configuration.

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Rebooting the system from the secondary operating system image, setting the secondary operating system image as the configured default boot image:

```
switch# boot system secondary
Default boot image set to secondary.

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? y
The system is going down for reboot.
```

Canceling a system reboot:

```
switch# boot system

Do you want to save the current configuration (y/n)? n

This will reboot the entire switch and render it unavailable
until the process is complete.
Continue (y/n)? n
Reboot aborted.
switch#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

show boot-history

show boot-history [all]

Description

Shows boot information. When no parameters are specified, shows the most recent information about the boot operation, and the three previous boot operations for the active management module. When the `all` parameter is specified, shows the boot information for the active management module and all available line modules. To view boot-history on the standby, the command must be sent on the standby console.

Parameter	Description
all	Shows boot information for the active management module and all available line modules.

Usage

This command displays the boot-index, boot-ID, and up time in seconds for the current boot. If there is a previous boot, it displays boot-index, boot-ID, reboot time (based on the time zone configured in the system) and reboot reasons. Previous boot information is displayed in reverse chronological order.

Index

The position of the boot in the history file. Range: 0 to 3.

Boot ID

A unique ID for the boot. A system-generated 128-bit string.

Current Boot, up for <SECONDS> seconds

For the current boot, the `show boot-history` command shows the number of seconds the module has been running on the current software.

Timestamp boot reason

For previous boot operations, the `show boot-history` command shows the time at which the operation occurred and the reason for the boot. The reason for the boot is one of the following values:

<DAEMON-NAME> crash

The daemon identified by <DAEMON-NAME> caused the module to boot.

Kernel crash

The operating system software associated with the module caused the module to boot.

Reboot requested through database

The reboot occurred because of a request made through the CLI or other API.

Uncontrolled reboot

The reason for the reboot is not known.

Examples

Showing the boot history of the active management module:

```
switch# show boot-history
Management module
=====

Index : 3
```

```

Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database
switch#

```

Showing the boot history of the active management module and all line modules:

```

switch# show boot-history all
Management module
=====

Index : 3
Boot ID : f1bf071bdd04492bbf8439c6e479d612
Current Boot, up for 22 hrs 12 mins 22 secs

Index : 2
Boot ID : edfa2d6598d24e989668306c4a56a06d
07 Aug 18 16:28:01 : Reboot requested through database

Index : 1
Boot ID : 0bda8d0361df4a7e8e3acdc1dba5caad
07 Aug 18 14:08:46 : Reboot requested through database

Index : 0
Boot ID : 23da2b0e26d048d7b3f4b6721b69c110
07 Aug 18 13:00:46 : Reboot requested through database

Line module 1/1
=====
Index : 3
10 Aug 17 12:45:46 : dune_agent crashed
...

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
All platforms	Manager (#)	Administrators or local user group members with execution rights for this command.

Chapter 6

Switch system and hardware commands

Switch system and hardware commands are general commands used to configure fundamental settings on the switch.



Refer to the Fundamentals Guide to view the switch system and hardware commands.

The switch has limited capacity to store data, collected by switch features and protocols. You can provide virtually unlimited storage capacity by adding user-supplied external storage volumes. Supported volume types and storage protocols include: NFSv3, NFSv4, and SCP (sshfs).

One application of external storage is the saving and restoring of DHCP lease files over SCP or NFS network attached storage systems. SCP file system protocol uses a user mode process to emulate a network file system. The key advantage is packet level encryption and simple configuration. The key disadvantage is slow performance.

You can set up external storage volume credentials and then enable it. A storage management process acts on your requests by enabling the storage volume using the requested storage protocol. You can disable the external storage volume or set it up but leave it disable.

The feature maintains storage volume state. The states are: **disabled** (down), **connecting** (establishing connection), **operational** (up), and **unaccessible** (unavailable).

If a storage volume is unavailable, the system attempts to reconnect periodically. Multiple volumes could connect concurrently. If one connection times out the others can connect immediately.

The system supports server connection through data and management ports.

Data port support requires server IP address on a default VRF.

Once a storage volume is enabled, applications can use the volume to store retrieve and delete files and directories.

External storage commands

address (external storage)

```
address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}  
no address {<IPV4-ADDR> | <IPV6-ADDR> | hostname <HOSTNAME>}
```

Description

Specifies the NAS IP address or hostname.

The `no` form of this command deletes an IP address or hostname.

Parameter	Description
<IPV4-ADDR>	Specifies the NAS server IPv4 address, Global.
<IPV6-ADDR>	Specifies the IPv6 address of the NAS server.
<HOSTNAME>	Specifies the hostname of the NAS server. String.

Examples

Creating the logfiles storage volume with IP address 10.1.1.1:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# address 10.1.1.1
```

Deleting an external storage volume named logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no address 10.1.1.1
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage- <i><VOLUME-NAME></i>	Administrators or local user group members with execution rights for this command.

directory

```
directory <DIRECTORY-NAME>  
no directory <DIRECTORY-NAME>
```

Description

Selects an existing directory on the external storage volume.

The `no` form of this command clears a directory of an external storage volume.

Parameter	Description
<i><DIRECTORY-NAME></i>	Specifies the external storage directory for mapping the volume.

Examples

Creating a volume named logfiles that is mapped under /home on the server:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# directory /home
```

Clearing the directory /home:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no directory /home
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage-<VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

disable external-storage logfiles

disable
no disable

Description

Disables the external storage volume.

The `no` form of this command enables the external storage volume. This is identical to the `enable` command.

Examples

Disabling a volume named logfiles:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage-<VOLUME-NAME>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only.

enable (external-storage logfiles)

enable
no enable

Description

Enables the external storage volume.

The `no` form of this command disables the external storage volume. This is identical to the `disable` command.

Examples

Creating and then enabling a volume named `logfiles`:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# enable
```

Disables the external storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# disable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	<code>config-external-storage-<i><VOLUME-NAME></i></code>	Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (<code>></code>) only.

external-storage

```
external-storage <VOLUME-NAME>
no external-storage <VOLUME-NAME>
```

Description

Creates or updates an external storage volume.

The `no` form of this command deletes an external storage volume.

Examples

Creating the `logfiles` storage volume:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)#
```

Deleting the `logfiles` storage volume:

```
switch(config)# no external-storage logfiles
```


Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config	Administrators or local user group members with execution rights for this command.

password (external-storage)

```
password [{plaintext | ciphertext} <PASSWORD>]  
no password {plaintext | ciphertext} <PASSWORD>
```

Description

Sets the password for network attached storage server login.

The `no` form of this command clears the password for network attached storage server login.

Parameter	Description
{ciphertext plaintext}	Selects the password format.
<PASSWORD>	Specifies the password. NOTE: When the password is not provided on the command line, plaintext password prompting occurs upon pressing Enter. The entered password characters are masked with asterisks.

Examples

Creating a volume named logfiles with password Xj#9:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# password plaintext Xj#9
```

Creating a volume named bak1 with a prompted plaintext password:

```
switch(config)# external-storage bak1  
switch(config-external-storage-bak1)# password  
Enter the NAS server password: *****  
Re-Enter the NAS server password: *****
```

Clearing the password for volume logfiles:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no password plaintext Xj#9
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

show external-storage

show external-storage [<VOLUME-NAME>]

Description

Shows external storage configuration and state for all volumes or for a specified volume.

Parameter	Description
<VOLUME-NAME>	Specifies the external storage volume name that the show command will use.

Examples

```
switch# show external-storage
```

```
-----  
--  
                Address      VRF      Username      Type      Directory      State  
-----  
--  
nfsvol    10.1.1.1      nas      ---           NFSv3      /home  
operational  
nfsfiles  20.1.1.1      nas      netstorage    NFSv4      /netstor      disabled  
scpdev    nasserver      nas      scpstor       SCP        /scp  
unaccessible
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320	Operator (>) or Manager	Administrators or local user group members with execution rights

Platforms	Command context	Authority
8325 8360	(#)	for this command.

show running-config external-storage

show running-config external-storage

Description

Shows the running configuration of the external storage.

Examples

```
switch# show running-config external-storage

external-storage nfsvol
  address 10.1.1.1
  vrf     nas
  type    nfsv4
  directoty /home
  enable
external-storage scpdev
  address 30.1.1.1
  vrf     nas
  username switchuser
  password ciphertext xxx
  type    scp
  directoty /home
  enable
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	Operator (>) or Manager (#)	Administrators or local user group members with execution rights for this command.

type (external storage)

type {nfsv3 | nfsv4 | scp}
no type {nfsv3 | nfsv4 | scp}

Description

Sets the network attached storage access type for reaching the external storage volume.

The **no** form of this command deletes an external storage volume.

Parameter	Description
nfsv3	Specifies the NFSv3 network access protocol.
nfsv4	Specifies the NFSv4 network access protocol.
scp	Specifies the SCP network access protocol.

Examples

Creating the logfiles volume using NFSV4:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# type nfsv4
```

Clearing the external storage access type:

```
switch(config)# external-storage logfiles
switch(config-external-storage-logfiles)# no type nfsv4
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

username (external storage)

```
username <USER-NAME>
no username <USER-NAME>
```

Description

Sets the username for logging in to a network attached storage server.

The `no` form of this command clears a username.

Parameter	Description
<USER-NAME>	Specifies the username.

Examples

Creating a volume named logfiles with the user name nassuser:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# username nasuser
```

Clearing the user name nasuser from accessing the logfiles volume:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no username nasuser
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

vrf (external storage)

```
vrf <VRF-NAME>  
no vrf <VRF-NAME>
```

Description

Setting a VRF to reach network attached storage.

The **no** form of this command clears access of a VRF to network attached storage.

Parameter	Description
<VRF-NAME>	Specifies the VRF name.

Examples

Creating the logfiles volume and setting a VRF named nas to access the network attached storage:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# vrf nas
```

Clearing access of a VRF named nas to the network attached storage:

```
switch(config)# external-storage logfiles  
switch(config-external-storage-logfiles)# no vrf nas
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-external-storage-<VOLUME-NAME>	Administrators or local user group members with execution rights for this command.

The IP Service Level Agreement (IP-SLA) is a feature that enables the measuring of network performance between two nodes in a network for different service level agreement parameters such as round-trip time (RTT), one-way delay, jitter, reachability, packet loss, and voice quality scores. These two nodes can span across area in access, distribution or core inside a LAN as well as across WAN between core to core or core to Data Centre switches. This feature helps you measure the SLA for different protocols or applications such as UDP echo, UDP jitter (for voice and video), TCP connect, HTTP, and ICMP echo. This guide provides details for managing and monitoring different types of IP-SLAs.

IP-SLA guidelines

- AOS-CX supports only SLA configuration through CLI and thresholds can be configured using NAE agents using WebUI/REST.
- AOS-CX supports only forever tests. On-demand tests are not supported.
- Maximum sessions: IP-SLA source 500, IP-SLA responder 100.
- NAE can effectively monitor a maximum of 300 parameters, reducing the maximum supported session by 300.
- NAE supports only syslog.
- NAE agents must be triggered for each IP-SLA test on every switch.
- If multiple IP addresses are received for a DNS query, DNS works with the first resolved IP.
- When the DNS server IP is not configured, the first DNS server in `resolve.conf` is used.
- The source interface/IP option is not applicable for SLAs configured on 'mgmt' VRF, as it has only one interface.
- A system time change because of NTP or a manual change causes an incorrect calculation.
- There is no interoperability of UDP echo SLA between AOS-CX and FlexFabric switches.
- Source IP and source port combination must be unique across SLA sessions in a same switch.
- Do not use the same source port across the source and responder sessions in a switch.
- NTP synchronization is a must for SLA types involving one-way delay such as UDP jitter VoIP.
- It is mandatory to set default CoPP to the max value when UDP jitter SLA is enabled otherwise 100% packet loss can be seen and `UDP-Jitter sla probe` will result in failure as seen in the following example.

```
copp-policy default
class hypertext priority 6 rate 50000 burst 64
default-class priority 6 rate 99999 burst 9999
```

- Deviations with respect to PVOS results: The packet losses due to internal switch-related issues like interface shutdown or interface flaps will not be considered as 'Probes Timed-out error', as the IP-SLA solution is to measure network performance and anomalies. Rather, this kind of packet loss will be counted in internal counters like 'Destination address unreachable'.

Limitations with VoIP SLAs

- A maximum of 80 concurrent VoIP SLAs can be scheduled in a 20 second slot.
- A single VoIP probe takes 20 seconds to complete.
- The default and minimum probe interval for VoIP SLA is 120 seconds.
- SLAs scheduled in the same slot, periodically sends 1000 probe packets for 120 seconds in 20 second intervals.
- Default 120 second probe interval is divided in to 6 slots of 20 seconds to avoid synchronization of all configured VoIP SLAs sending probes at the same time.
- SLAs started at the same time exceeding the concurrent limit of 80 must wait for the next 20 second VoIP slot to open before moving to 'running' state.
- The maximum number of VoIP SLAs supported is 80 X 6 slots = 480 SLAs.
- SLAs exceeding 480 will continue to remain in the 'waiting for VoIP slot' until any slot is freed by stopping the running SLA.
- To avoid high RTT, a single switch with more than 20 SLAs should not have single responder SLA.
- When IP is received dynamically (e.g. using DHCP) for interfaces other than management interface, IPSLA source or responder has to be configured only using interface name.

IP-SLA commands

http

```
http {get | raw} URL [source {<SOURCE-IPV4-ADDR> | <IFNAME>} source-port <PORT-NUM>]  
[proxy proxy-url] [cache disable] [name-server <IPV4-ADDR-DNS-SERVER>]  
[probe-interval <30-604800>] [version<VERSION-NUMBER>] [http-raw-request <RAW-  
PAYLOAD>]
```

Description

Configures HTTP as the IP-SLA test mechanism. Requires destination URL and type of HTTP request (raw/get).

Parameter	Description
{get raw}	Selects HTTP request type as GET or RAW where the system will generate or provide HTTP payload.
URL	Specifies HTTP URL address of syntax. http://<HOST NAME/IP-ADDRESS>:<PORT>/<PATH>.
source {<SOURCE-IPV4-ADDR> <IFNAME>}	Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.
source-port <PORT-NUM>	Specifies the value of the source port for the IP-SLA probes.
cache disable	Selects cache option for the HTTP server. By default the option is enabled.
name-server <IPV4-ADDR-DNS-SERVER>	Specifies the IPv4 address of DNS server.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 30 to 604800.

Parameter	Description
version <VERSION-NUMBER>	Specifies the source interface to use for sending IP-SLA probes.
http-raw-request <RAW-PAYLOAD>	HTTP raw request. String.

Examples

```
switch(config-ipsla-1)# http get http://device.arubanetworks.com/root/home.html
switch(config-ipsla-1)# http raw
http://device.arubanetworks.com/root/home.html
switch(config-ipsla-1)# http 2.2.2.2 source 1/1/1
switch(config-ipsla-1)# http http://device.arubanetworks.com source 2.2.2.1
switch(config-ipsla-1)# http http://device.arubanetworks.com/root/home.html
source-interface 1/1/1
switch(config-ipsla-1)# http http://device.arubanetworks.com name-server
10.10.10.2
switch(config-ipsla-1)# http raw raw-request "GET /en/US/hmpgs/index.html
HTTP/1.0\r\n\r\n"
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

icmp-echo

```
icmp-echo {<DEST-IPV4-ADDR>|<HOSTNAME>} [source {<SOURCE-IPV4-ADDR> | <IFNAME>}]
[name-server <IPV4-ADDR-DNS-SERVER>] [payload-size <PAYLOAD-SIZE>]
[tos <TYPE-OF-SERVICE>] [probe-interval <PROBE-INTERVAL>]
```

Description

Configures ICMP echo as the IP-SLA test mechanism. Requires destination address for the IP-SLA test.

Parameter	Description
{<DEST-IPV4-ADDR> <HOSTNAME>}	Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.
[source {<SOURCE-IPV4-ADDR> <IFNAME>}]	Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.
name-server <IPV4-ADDR-DNS-SERVER>	Specifies the DNS server for destination hostname resolution.

Parameter	Description
payload-size <PAYLOAD-SIZE>	Specifies the payload size of an SLA probe. Range: 0 to 1440.
tos <TYPE-OF-SERVICE>	Specifies the type of serve to be used in the probe packets. Range: 0 to 255.
probe-interval <PROBE-INTERVAL>	Specifies the probe interval in seconds. Range: 5 to 604800.

Examples

```
switch(config)# ip-sla test
  switch(config-ip-sla-test)# icmp-echo 2.2.2.2
  switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3
  switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3 payload-size 400
  switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3 payload-size 400
name-server 4.4.4.4
  switch(config-ip-sla-test)# icmp-echo 2.2.2.2 source 3.3.3.3 payload-size 400
name-server 4.4.4.4 probe-interval 80
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config-ip-sla-<IP-SLA-NAME>	Administrators or local user group members with execution rights for this command.

ip-sla

```
ip-sla <IP-SLA-NAME>
no ip-sla <IP-SLA-NAME>
```

Description

Creates an IP Service Level Agreement (SLA) profile and switches to the `config-ip-sla` context.

The `no` form of this command deletes an IP-SLA profile. By default, all profile use the default VRF (default).

Parameter	Description
<IP-SLA-NAME>	Specifies an IP-SLA profile name. Length: 1 to 63 characters.

Examples

Creating an IP-SLA:

```
switch(config)# ip-sla 1
switch(config-ip-sla-1)#
```

Deleting an IP-SLA:

```
switch(config)# no ip-sla 1
switch(config)#
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config	Administrators or local user group members with execution rights for this command.

ip-sla responder

```
ip-sla responder <SLA-NAME> {udp-echo | tcp-connect | udp-jitter-voip} <PORT-NUM>
[source {<SOURCE-IPV4-ADDR> | <IFNAME>}] [vrf <VRF-NAME>]
no ip-sla responder <SLA-NAME> {udp-echo | tcp-connect | udp-jitter-voip} <PORT-NUM>
[source {<SOURCE-IPV4-ADDR> | <IFNAME>}] [vrf <VRF-NAME>]
```

Description

Selects the IP-SLA responder. The responder can be configured for udp-echo, tcp-connect, udp-jitter-voip type. It requires the SLA name, SLA type, and port number as arguments. Source IP/interface ID is a must for type udp-jitter-voip and optional for other types.

The **no** form of this command removes the IP-SLA responder.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
udp-echo	Enables responder for udp-echo probes.
tcp-connect	Selects TCP connect as the IP-SLA test mechanism.
vrf <VRF-NAME>	Specifies the name of the VRF to use.
udp-jitter-voip	Selects VOIP jitter as the IP-SLA test mechanism.
<PORT-NUM>	Specifies the port number to listen for IP-SLA probes. Range: 1 to 65535.
[source {<SOURCE-IPV4-ADDR> <IFNAME>}]	Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes.

Examples

```
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
switch(config)# ip-sla responder SLA1 udp-echo 8000 source 1/1/1
```

```
switch(config)# no ip-sla responder SLA1 udp-echo 8000 source 2.2.2.2
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config	Administrators or local user group members with execution rights for this command.

show ip-sla responder

show ip-sla responder <SLA-NAME>

Description

Shows the given IP-SLA responder configuration and operation status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.

Examples

```
switch(config)# show ip-sla responder SLA3

SLA Name       : SLA3
IP-SLA Type    : Udp-echo
VRF            : Default
Responder Port  : 8000
Responder IP    : 2.2.2.3
Responder Interface : 1/1/1
Responder Status : Running
```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config	Administrators or local user group members with execution rights for this command.

show ip-sla responder results

show ip-sla responder <SLA-NAME> <SOURCE-IPV4-ADDR> <PORT-NUM> results

Description

Shows the given ip-sla responder statistics for a given source IP and port. This command is only applicable for the sources where source IP and port are configured.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
<SOURCE-IPV4-ADDR>	Specifies the source IPV4 address.
<PORT-NUM>	Specifies the port number. Range: 1 to 65535.

Examples

```
switch# show ip-sla responder SLA1 2.2.2.1 8000 results
```

```

IP-SLA Type       : Udp-echo
VRF Name          : Default
Source IP         : 2.2.2.1
Source Port       : 8000
Responder Port    : 8888
Responder IP      : 2.2.2.3
Responder Interface :
Responder Status  : Running
Packets Received  : 2
Packets Sent      : 2

```

Command History

Release	Modification
10.07 or earlier	--

Command Information

Platforms	Command context	Authority
8320 8325 8360	config	Administrators or local user group members with execution rights for this command.

show ip-sla <SLA-NAME>

show ip-sla <SLA-NAME> results

Description

Shows the given IP-SLA source configuration and status.

Parameter	Description
<SLA-NAME>	Specifies the SLA name.
results	Shows the statistics calculated for an SLA type.

Examples

```
switch# show ip-sla xyz results

IP-SLA session status
  IP-SLA Name           : xyz
  IP-SLA Type           : tcp-connect
  Destination Host Name/IP Address: 2.2.2.1
  Destination Port       : 8888
  Source IP Address/IFName : 2.2.2.2
  Source Port           : 5555
  Status                : Running

IP-SLA session cumulative counters
  Total Probes Transmitted : 1
  Probes Timed-out        : 0
  Bind Error               : 0
  Destination Address Unreachable : 0
  DNS Resolution Failures : 0
  Reception Error         : 0
  Transmission Error       : 0

IP-SLA Latest Probe Results
  Last Probe Time        : 2018 Jul 13 02:00:35
  Packets Sent           : 1
  Packets Received       : 1
  Packet Loss in Test    : 0.0000%

  Minimum RTT(ms)        : 0.7900
  Maximum RTT(ms)        : 0.7900
  Average RTT(ms)        : 0.7900
  DNS RTT(ms)            : 0.0000
  TCP RTT(ms)            : 0.9710

switch(config)# show ip-sla xyz
  IP-SLA Name           : xyz
  Status                : scheduled
  IP-SLA Type           : tcp-connect
  VRF                   : ipslasrc
  Source Port           : 5555
  Source IP             : 2.2.2.2
  Source Interface      :
  Domain Name Server    :
  Probe interval(seconds) : 90

switch(config)# show ip-sla jitter-sla results
  IP-SLA session status
    IP-SLA Name           : jitter-sla
    IP-SLA Type           : udp-jitter-voip
    Destination Host Name/IP Address: 2.2.2.1
```

```

Destination Port      : 8888
Source IP Address/IFName :
Source Port          : 5555
Status               : Running

IP-SLA Session Cumulative Counters
Total Probes Transmitted : 1
Probes Timed-out        : 0
Bind Error              : 0
Destination Address Unreachable : 0
DNS Resolution Failures : 0
Reception Error         : 0
Transmission Error      : 0

IP-SLA Latest Probe Results
Last Probe Time        : 2018 Jul 13 02:02:48
Packets Sent           : 1
Packets Received       : 1
Packet Loss in Test    : 0.0000%

Minimum RTT(ms)        : 0.7900
Maximum RTT(ms)        : 0.7900
Average RTT(ms)        : 0.7900
DNS RTT(ms)            : 0.0000

Min Positive SD         : 1      Min Positive DS         : 2
Max Positive SD         : 1      Max Positive DS         : 2
Positive SD Number      : 2      Positive DS Number      : 2
Positive SD Sum         : 2      Positive DS Sum         : 4
Positive SD Average     : 5      Positive DS Average     : 5
Min Negative SD         : 1      Min Negative DS         : 1
Max Negative SD         : 1      Max Negative DS         : 1
Negative SD Number      : 2      Negative DS Number      : 4
Negative SD Sum         : 2      Negative DS Sum         : 4
Negative SD Average     : 5      Negative DS Average     : 5

Max SD Delay            : 0      Max DS Delay            : 0
Min SD Delay            : 0      Min DS Delay            : 0
Average SD Delay        : 0      Average DS Delay        : 0

Voice Scores:
MOS   Score            : 4.38    ICPIF                   : 0

```

```

switch(config)# show ip-sla m3op
IP-SLA Name           : jitter-sla
Status                 : Running
IP-SLA Type            : udp-jitter-voip
VRF                    : ipslasrc
Source IP              : 2.2.2.2
Source Interface       :
Domain Name Server     :
TOS                    : 10
Probe Interval(seconds) : 90
Advantage Factor       : 0
Codec Type             : g711a

```

```

switch(config)# show ip-sla http-sla
IP-SLA Name           : http-sla
Status                 : Running
IP-SLA Type            : http

```

```

VRF                : ipslasrc
Source IP           : 2.2.2.2
Source Interface    :
Domain Name Server  : 10.10.10.2
Probe Interval(seconds) : 90
HTTP Request Type   : GET
HTTP/HTTPS URL      : abcd.com/ws/home
Cache               : Enabled
HTTP Proxy URL      :
HTTP Version Number : 1.1
```

IP-SLA status description
```
| Status | Description |
|-----|-----|
| Running | SLA is fully operational |
| Bind Error | Another service is using the same source port |
| Interface Down | Interface status is not up |
| Dns Resolution Error | Failed to resolve destination hostname |
| No Route | No available route to the responder |
| Internal Error | Unexpected error prevents SLA session |
| Disabled | SLA is disabled |
| Configuration Incomplete | Configuration is not complete to enable the SLA |
```

IP SLA session cumulative counters description
```
| Status | Description |
|-----|-----|
| Probes Timed-out response. | Total numbers of probes failed to receive response. |
| Bind Error as source port not available. | Total numbers of probes transmission failed as source port not available. |
| Destination Address Unreachable due to route unavailable. | Total numbers of probes transmission failed due to route unavailable. |
| DNS Resolution Failures resolution failure. | Total numbers of probes failed due to DNS resolution failure. |
| Reception Error internal error in reception. | Total numbers of probes failed due to internal error in reception. |
| Transmission Error internal errr in transmission. | Total numbers of probes failed due to internal errr in transmission. |
```

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | Operator (>) or Manager (#) | Administrators or local user group members with execution rights for this command. |



## start-test

start-test

### Description

Starts the IP-SLA probes.

### Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# start-test
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms            | Command context                           | Authority                                                                          |
|----------------------|-------------------------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-ip-sla- <i>&lt;IP-SLA-NAME&gt;</i> | Administrators or local user group members with execution rights for this command. |

## stop-test

stop-test

### Description

Stops the IP-SLA probes.

### Examples

```
switch(config)# ip-sla test
switch(config-ip-sla-test)# stop-test
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms            | Command context                           | Authority                                                                          |
|----------------------|-------------------------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-ip-sla- <i>&lt;IP-SLA-NAME&gt;</i> | Administrators or local user group members with execution rights for this command. |

## tcp-connect

```
tcp-connect {<DEST-IPV4-ADDR> | <HOSTNAME>} <PORT-NUM> [source {<SOURCE-IPV4-ADDR> |
<IFNAME>} [source-port <PORT-NUM>]] [name-server <IPV4-ADDR-DNS-SERVER>]
[probe-interval <PROBE-INTERVAL>]
```

### Description

Configures TCP connect as the IP-SLA test mechanism. Requires destination address/hostname and destination port for the IP-SLA of tcp-connect IP-SLA type.

| Parameter                                | Description                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| {<DEST-IPV4-ADDR>   <HOSTNAME>}          | Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.                  |
| <PORT-NUM>                               | Destination port for the IP-SLA. Range: 1 to 65535.                                                      |
| [source {<SOURCE-IPV4-ADDR>   <IFNAME>}] | Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes. |
| [source-port <PORT-NUM>]                 | Specifies the port for the IP-SLA test.                                                                  |
| [name-server <IPV4-ADDR-DNS-SERVER>]     | Specifies the DNS server for destination hostname resolution.                                            |
| [probe-interval <PROBE-INTERVAL>]        | Probe interval in seconds. Range: 30 to 604800.                                                          |

### Examples

```
switch(config-ipsla-1)# tcp-connect 2.2.2.2 8080
switch(config-ipsla-1)# tcp-connect 2.2.2.2 8080 source 2.2.2.1 source-port
6000
switch(config-ipsla-1)# tcp-connect 2.2.2.2 8080 source 1/1/1 source-port
6000

switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
source 2.2.2.1 source-port 6000
switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
source 1/1/1 source-port 6000
switch(config-ipsla-1)# tcp-connect https://device.arubanetworks.com 8080
name-server 10.10.10.2
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms            | Command context             | Authority                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-ip-sla-<IP-SLA-NAME> | Administrators or local user group members with execution rights for this command. |

## udp-echo

```
udp-echo {<DEST-IPV4-ADDR>|<HOSTNAME>} <PORT-NUM> [source {<SOURCE-IPV4-ADDR> |
<IFNAME>} [source-port <PORT-NUM>]] [name-server <IPV4-ADDR-DNS-SERVER>] [payload-
size
 <PAYLOAD-SIZE>] [tos <TYPE-OF-SERVICE>] [probe-interval <PROBE-INTERVAL>]
```

### Description

Configures UDP echo as the IP-SLA test mechanism. Requires destination address/hostname and destination port number for the IP-SLA of udp-echo SLA type.

| Parameter                                | Description                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| {<DEST-IPV4-ADDR>   <HOSTNAME>}          | Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.                  |
| <PORT-NUM>                               | Specifies the destination port for the IP-SLA. Range: 1 to 65535.                                        |
| [source {<SOURCE-IPV4-ADDR>   <IFNAME>}] | Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes. |
| [source-port <PORT-NUM>]                 | Specifies source port for the IP-SLA test. Range: 1 to 65535.                                            |
| [name-server <IPV4-ADDR-DNS-SERVER>]     | Specifies the DNS server for destination hostname resolution.                                            |
| [payload-size <PAYLOAD-SIZE>]            | Specifies the payload size of an SLA probe. Range: 28 to 1440.                                           |
| [<TYPE-OF-SERVICE>]                      | Type of service. Range: 0 to 255.                                                                        |
| probe-interval <PROBE-INTERVAL>          | Probe interval in seconds. Range: 5 to 604800.                                                           |

### Examples

```
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 2.2.2.1
switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 1/1/1
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 2.2.2.1 payload-size 50
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 source 1/1/1 payload-size 50
switch(config-ipsla-1)# udp-echo 2.2.2.2 8080 payload-size 50
switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080 source
2.2.2.1
 payload-size 50
switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080 source
1/1/1
 payload-size 50
switch(config-ipsla-1)# udp-echo https://device.arubanetworks.com 8080
 name-server 10.10.10.2
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-ip-sla-<IP-SLA-NAME> | Administrators or local user group members with execution rights for this command. |

## udp-jitter-voip

```
udp-jitter-voip {<DEST-IPV4-ADDR> | <HOSTNAME>} <PORT-NUM> [codec-type <CODEC-TYPE>]
[advantage-factor <VALUE>] [source {<SOURCE-IPV4-ADDR> | <IFNAME>} [source-port
<PORT-NUM>]]
[name-server <IPV4-ADDR-DNS-SERVER>] [probe-interval <PROBE-INTERVAL>] [tos <TYPE-OF-
SERVICE>]
```

## Description

Configure UDP jitter voip as the IP-SLA test mechanism. Requires destination address/hostname and source address/interface for the IP-SLA of udp-jitter-voip IP-SLA type.

| Parameter                                | Description                                                                                              |
|------------------------------------------|----------------------------------------------------------------------------------------------------------|
| {<DEST-IPV4-ADDR>   <HOSTNAME>}          | Selects the destination IPv4 address for the IP-SLA or the hostname of the destination.                  |
| <PORT-NUM>                               | Selects the port number for the IP-SLA. Range: 1 to 65535.                                               |
| [codec-type <CODEC-TYPE>]                | Selects the codec-type for the Voip IP-SLA test.                                                         |
| [advantage-factor <ADVANTAGE-FACTOR>]    | Selects the value for the advantage factor. Default value is 0.                                          |
| [source {<SOURCE-IPV4-ADDR>   <IFNAME>}] | Selects the source IPv4 address for SLA probes or the source interface to use for sending IP-SLA probes. |
| [source-port <PORT-NUM>]                 | Specifies the value of source port for the IP-SLA probes.                                                |
| [name-server <IPV4-ADDR-DNS-SERVER>]     | Specifies the DNS server for destination hostname resolution.                                            |
| tos <TYPE-OF-SERVICE>                    | Specifies the type of service. Range: 0 to 255.                                                          |
| probe-interval <PROBE-INTERVAL>          | Specifies the probe interval in seconds. Range: 120 to 604800.                                           |

## Examples

```
switch(config-ipsla-1)# udp-jitter-voip 2.2.2.2 8080 advantage-factor 10 codec-
type g711a
switch(config-ipsla-1)# udp-jitter-voip 2.2.2.2 8080 advantage-factor 10
codec-type g711a source 2.2.2.1
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a
switch(config-ipsla-1)# udp-jitter-voip 2.2.2.2 8080 advantage-factor 10
```

```

codec-type g711a source 1/1/1
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a source 2.2.2.1
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a source 1/1/1
switch(config-ipsla-1)# udp-jitter-voip https://device.arubanetworks.com 8080
advantage-factor 10 codec-type g711a name-server 10.10.10.2 probe-interval 120
source 10.1.1.1 source-port 8888 tos 10

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context             | Authority                                                                          |
|----------------------|-----------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-ip-sla-<IP-SLA-NAME> | Administrators or local user group members with execution rights for this command. |

## vrf (ip sla)

```

vrf <VRF-NAME>
no vrf [<VRF-NAME>]

```

## Description

Configures the VRF on which the SLA will send or receive packets. By default, the default VRF is used. The `no` form of the command removes VRF from SLA.

| Parameter  | Description                                     |
|------------|-------------------------------------------------|
| <VRF-NAME> | Specifies a VRF name. Length: Default: default. |

## Examples

```
switch(config-ip-sla-test)# vrf ipslasrc
```

```
switch(config-ip-sla-test)# no vrf
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context                                | Authority                                                                          |
|----------------------|------------------------------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | <code>config-ip-sla-&lt;IP-SLA-NAME&gt;</code> | Administrators or local user group members with execution rights for this command. |

## L1-100Mbps downshift commands

### downshift enable

```
downshift-enable
no downshift-enable
```

#### Description

Enables/disables automatic speed downshift on an interface that supports downshift, generally 1GBASE-T ports. When enabled, downshift allows an interface to link at a lower advertised speed when unable to establish a stable link at the maximum speed. Downshifting only applies to physical interfaces that are not members of a LAG and is only available when auto-negotiation is enabled. When only one speed is advertised, downshift will not be triggered.

#### Examples

```
switch(config-if)# interface 1/1/1
switch(config-if)# downshift-enable
```

Warning: this is a non-standard mode for use only when standards-based auto-negotiation is not able to establish a stable link. Enabling this may cause the port to link at a lower than expected speed and should not be used on ports that are members of a LAG. Support calls may require this feature to be disabled

Continue (y/n)?

```
switch(config-if)#
```

When automatic downshift is enabled:

```
switch(config-if)# show running-config interface
interface 1/1/1
 downshift-enable
```

Disabling automatic speed downshift:

```
switch(config-if)# interface 1/1/1
switch(config-if)# no downshift-enable
```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## show interface

```
show interface [<IFNAME>|<IFRANGE>] [brief | physical | extended [non-zero]]
show interface [lag | loopback | tunnel | vlan] [<ID>] [brief | physical]
show interface [lag | loopback | tunnel | vlan] [<ID>] [extended [non-zero]]
show interface vxlan <ID> [brief | physical]
show interface vxlan <ID> [brief | physical]
```

## Description

Shows active configurations and operational status information for interfaces.

| Parameter     | Description                                           |
|---------------|-------------------------------------------------------|
| <IFNAME>      | Specifies a interface name.                           |
| <IFRANGE>     | Specifies the port identifier range.                  |
| brief         | Shows brief info in tabular format.                   |
| physical      | Shows the physical connection info in tabular format. |
| extended      | Shows additional statistics.                          |
| non-zero      | Shows only non zero statistics.                       |
| LAG           | Shows LAG interface information.                      |
| LOOPBACK      | Shows loopback interface information.                 |
| TUNNEL        | Shows tunnel interface information.                   |
| VLAN          | Shows VLAN interface information.                     |
| <LAG-ID>      | Specifies the LAG number. Range: 1-256                |
| <LOOPBACK-ID> | Specifies the LOOPBACK number. Range: 0-255           |
| <TUNNEL-ID>   | Specifies the tunnel ID. Range: 1-255                 |
| <VLAN-ID>     | Specifies the VLAN ID. Range: 1-4094                  |
| VXLAN         | Shows the VXLAN interface information.                |
| <VXLAN-ID>    | Specifies the VXLAN interface identifier. Default: 1  |

## Examples

The following example shows when the interface is configured as a route-only port:

```
switch# show interface 1/1/1

Interface 1/1/1 is up
Admin state is up
Link state: up for 2 days (since Sun Jun 21 05:30:22 UTC 2020)
Link transitions: 1
Description: backup data center link

Hardware: Ethernet, MAC Address: 70:72:cf:fd:e7:b4
MTU 1500
Type 1GbT
Full-duplex
qos trust none
Speed 1000 Mb/s
Auto-negotiation is on
Flow-control: off
Error-control: off
MDI mode: MDIX
L3 Counters: Rx Enabled, Tx Enabled
Rate collection interval: 300 seconds
```

| Rates         | RX   | TX   | Total (RX+TX) |
|---------------|------|------|---------------|
| Mbits / sec   | 0.00 | 0.00 | 0.00          |
| KPkts / sec   | 0.00 | 0.00 | 0.00          |
| Unicast       | 0.00 | 0.00 | 0.00          |
| Multicast     | 0.00 | 0.00 | 0.00          |
| Broadcast     | 0.00 | 0.00 | 0.00          |
| Utilization % | 0.00 | 0.00 | 0.00          |

| Statistics   | RX  | TX  | Total |
|--------------|-----|-----|-------|
| Packets      | 0   | 0   | 0     |
| Unicast      | 0   | 0   | 0     |
| Multicast    | 0   | 0   | 0     |
| Broadcast    | 0   | 0   | 0     |
| Bytes        | 0   | 0   | 0     |
| Jumbos       | 0   | 0   | 0     |
| Dropped      | 0   | 0   | 0     |
| Filtered     | 0   | 0   | 0     |
| Pause Frames | 0   | 0   | 0     |
| L3 Packets   | 0   | 0   | 0     |
| L3 Bytes     | 0   | 0   | 0     |
| Errors       | 0   | 0   | 0     |
| CRC/FCS      | 0   | n/a | 0     |
| Collision    | n/a | 0   | 0     |
| Runts        | 0   | n/a | 0     |
| Giants       | 0   | n/a | 0     |
| Other        | 0   | 0   | 0     |

When the interface is currently linked at a downshifted speed:

```
switch(config-if)# show interface 1/1/1

Interface 1/1/1 is up
...
Auto-negotiation is on with downshift active
```

When the interface is shut down during a VSX split:



```
switch(config-if)# show interface 1/1/1
```

```
Interface 1/1/1 is down
Admin state is up
State information: Disabled by VSX
Link state: down for 3 days (since Tue Mar 16 05:20:47 UTC 2021)
Link transitions: 0
Description:
```

```
Hardware: Ethernet, MAC Address: 04:09:73:62:90:e7
MTU 1500
Type SFP+DAC3
Full-duplex
qos trust none
Speed 0 Mb/s
Auto-negotiation is off
Flow-control: off
Error-control: off
VLAN Mode: native-untagged
Native VLAN: 1
Allowed VLAN List: 1502-1505
Rate collection interval: 300 seconds
```

| Rate        | RX   | TX   | Total (RX+TX) |
|-------------|------|------|---------------|
| Mbits / sec | 0.00 | 0.00 | 0.00          |
| KPkts / sec | 0.00 | 0.00 | 0.00          |
| Unicast     | 0.00 | 0.00 | 0.00          |
| Multicast   | 0.00 | 0.00 | 0.00          |
| Broadcast   | 0.00 | 0.00 | 0.00          |
| Utilization | 0.00 | 0.00 | 0.00          |

| Statistic    | RX  | TX  | Total |
|--------------|-----|-----|-------|
| Packets      | 0   | 0   | 0     |
| Unicast      | 0   | 0   | 0     |
| Multicast    | 0   | 0   | 0     |
| Broadcast    | 0   | 0   | 0     |
| Bytes        | 0   | 0   | 0     |
| Jumbos       | 0   | 0   | 0     |
| Dropped      | 0   | 0   | 0     |
| Pause Frames | 0   | 0   | 0     |
| Errors       | 0   | 0   | 0     |
| CRC/FCS      | 0   | n/a | 0     |
| Collision    | n/a | 0   | 0     |
| Runts        | 0   | n/a | 0     |
| Giants       | 0   | n/a | 0     |

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context         | Authority                                                    |
|---------------|-------------------------|--------------------------------------------------------------|
| All platforms | Operator (>) or Manager | Operators or Administrators or local user group members with |

| Platforms | Command context | Authority                                                                                                 |
|-----------|-----------------|-----------------------------------------------------------------------------------------------------------|
|           | (#)             | execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show interface downshift-enable

```
show interface [<IFNAME>|<IFRANGE>] downshift-enable
```

### Description

Displays speed downshift information, including the interface speed status and configuration.

| Parameter | Description                          |
|-----------|--------------------------------------|
| <IFNAME>  | Specifies a interface name.          |
| <IFRANGE> | Specifies the port identifier range. |

### Examples

Showing automatic downshift information:

```
switch(config-if)# show interface downshift-enable

Port Downshift Speed
 Enabled | Active Status | Config

1/1/1 yes yes 100M-FDx auto
1/1/2 yes no 1G auto
1/1/3 yes no 100M-FDx 100M-FDx
1/1/4 no no -- auto
```

Showing automatic downshift information on per interface:

```
switch(config-if)# show interface 1/1/2 downshift-enable

Port Downshift Speed
 Enabled | Active Status | Config

1/1/2 yes no 1G auto
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                                                                                                              |
|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show running-config interface

```
show running-config interface [<IFNAME>|<IFRANGE>]
show running-config interface [lag | loopback | tunnel | vlan] [<ID>]
show running-config interface vxlan <ID>
```

### Description

Displays active configurations of various switch interfaces.

| Parameter     | Description                                           |
|---------------|-------------------------------------------------------|
| <IFNAME>      | Specifies a interface name.                           |
| <IFRANGE>     | Specifies the port identifier range.                  |
| LAG           | Specifies LAG interface information                   |
| LOOPBACK      | Specifies loopback interface information.             |
| TUNNEL        | Specifies tunnel interface information.               |
| VLAN          | Specifies VLAN interface information.                 |
| <LAG-ID>      | Specifies the LAG number. Range: 1-256.               |
| <LOOPBACK-ID> | Specifies the LOOPBACK number. Range: 0-255.          |
| <TUNNEL-ID>   | Specifies the tunnel ID. Range: 1-255.                |
| <VLAN-ID>     | Specifies the VLAN ID. Range: 1-4094.                 |
| VXLAN         | Specifies the VXLAN interface information.            |
| <VXLAN-ID>    | Specifies the VXLAN interface identifier. Default: 1. |

### Examples

Showing 1/1/2 interface configuration:

```
switch(config-if)# show running-config interface 1/1/2

interface 1/1/2
 no shutdown
 description DC-23
exit
```

Showing loopback interfaces configured:

```
switch(config-if)# show running-config interface loopback

interface loopback 1
 description lb interface 1
 exit
interface loopback 2
 description lb interface 2
 exit
```

Showing loopback interfaces not configured:

```
switch(config-if)# show running-config interface loopback

No loopback interfaces configured.
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                                                                                                              |
|-----------|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | config          | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

Mirroring allows you to replicate all traffic arriving and/or leaving the selected system interfaces. This data can be used for collection or analysis.

The traffic replicated using mirroring can be sent to a separate interface on the same switch as the traffic source for analysis or inspection. Such a collection of interfaces and settings is called a mirror session.

A mirror session can be configured with many traffic sources but only a single output, or destination. In the initial configuration, the mirror session is disabled. You have enable the feature to start the replication.



---

Care must be taken in choosing the number and rates of sources to avoid over-saturating a session destination. A mirror session with multiple 10G sources can overwhelm a single 10G destination and important data may be lost.

---

## Mirroring statistics and sFlow

Mirror statistics are reset for a mirror-to-CPU session when an interface is added or removed from a LAG that is a source interface in the mirror session.

Mirroring and sFlow configuration on the same port is supported.

## Limitations

The following limitations apply when configuring multiple mirroring sessions on a switch:

- CPU generated packets egressing on a routed L3 interface will not be mirrored to the destination port.
- Untagged egress packets that get mirrored will have the native VLAN tag in the mirrored packet. These extra bytes can cause traffic loss at the mirror destination when running line rate traffic.
- True egress mirroring is not supported on 832x platforms. Egress mirroring takes place at the ingress. The packets that may get dropped at the egress might also have been mirrored at ingress. Traffic will be mirrored even before the policy actions are processed at the egress.
- Packets mirrored to CPU from a Layer-3 Route Only Port (ROP) will have a VLAN tag with the VLAN ID set to the internal VLAN ID assigned to that port.
- 832x platforms have 4 mirror ASIC resources that can be used among the different mirror sessions. Each direction in a mirror session will consume 1 mirror ASIC resource. Hence, a user can have up to 4 unidirectional mirror sessions or 2 bi-directional mirror sessions active at any given time. If there are no mirror ASIC resources available when attempting to enable a mirror session, the 'Operation Status' field of `show mirror` command for session ID will have the status set to 'platform\_session\_limit\_reached.'

- The mirror destination port among the active mirror sessions must be unique i.e. if an interface is configured as a source or destination in an active mirror session, the same port cannot be used as a destination in another active mirror session.
- The interface/LAG used to transmit ERSPAN packets cannot be a source in *any* mirror session.
- The interface/LAG used to transmit ERSPAN packets must be unique per ERSPAN mirror session. If a change in the L3 topology causes multiple ERSPAN mirror sessions to use the same egress interface/LAG to transmit the ERSPAN packets, then only one session will work. The other session(s) will go into an error state.

## Mirroring commands

### clear mirror

```
clear mirror [all | <SESSION-ID>]
```

#### Description

Clears the mirror statistics for all configured mirror sessions or a specified session

| Parameter    | Description                                                   |
|--------------|---------------------------------------------------------------|
| all          | Specifies all configured sessions.                            |
| <SESSION-ID> | Specifies a numeric identifier for the session. Range: 1 to 4 |

#### Examples

Clearing mirror statistics for all configured mirror sessions:

```
switch# clear mirror all
```

Clearing mirror statistics for mirror session 1:

```
switch# clear mirror 1
```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

#### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Manager (#)     | Administrators or local user group members with execution rights for this command. |

### comment

```
comment <COMMENT>
```

no comment

## Description

Specifies a comment for the mirroring session.

When used in mirror endpoint command context, specifies a comment for the mirror endpoint.

The `no` form of this command removes the comment.

| Parameter                    | Description                                                                                                     |
|------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>&lt;COMMENT&gt;</code> | A comment string of up to 64 characters composed of letters, numbers, underscores, dashes, spaces, and periods. |

## Usage

Comments are optional and can be added or removed at any time without affecting the state of the mirroring session.

Adding a comment to a session that already has a comment replaces the existing comment.

## Examples

Adding a comment to a mirror session:

```
switch(config-mirror-3) # comment This Mirror will be removed during next
maintenance window
```

Removing the comment from mirror session 3:

```
switch(config-mirror-3) # no comment
```

Adding a comment to a mirror endpoint:

```
switch(config-mirror-endpoint-test) # comment Monitor endpoint traffic
```

Replacing the existing comment for mirror endpoint:

```
switch(config-mirror-endpoint-test) # comment Monitor statistics on each endpoint
interfaces
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context                                      | Authority                                                                          |
|---------------|------------------------------------------------------|------------------------------------------------------------------------------------|
| All platforms | config-mirror-<SESSION-ID><br>config-mirror-endpoint | Administrators or local user group members with execution rights for this command. |

## copy tcpdump-pcap

copy tcpdump-pcap <FILE-NAME> <REMOTE-URL>

### Description

Saves packet capture files to external storage.

| Parameter    | Description                                                                    |
|--------------|--------------------------------------------------------------------------------|
| <FILE-NAME>  | Specifies the packet capture file to save.                                     |
| <REMOTE-URL> | Specifies the external storage to which the packet capture file will be saved. |

### Usage

Only four files can be saved at any point on the switch. Packet capture files are not saved after a failover or reboot. View a list of saved files using `diag utilities list-files`.

### Examples

Saving my\_capture\_file.pcap to sftp://root@10.0.0.2/file.pcap:

```
switch# copy tcpdump-pcap my_capture_file.pcap sftp://root@10.0.0.2/file.pcap
root@10.0.0.2's passowrd:
Connected to 10.0.0.2.
sftp > put my_capture_file.pcap file.pcap
Uploading my_capture_file.pcap to /root/file.pcap
my_capture_file.pcap 100% 156 219.8KB/s 00:00
Copied successfully.
```

### Command History

| Release | Modification       |
|---------|--------------------|
| 10.08   | Command introduced |

### Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## copy tshark-pcap

copy tshark-pcap <REMOTE-URL> [vrf <VRF-NAME>]



## Description

Copies the tshark capture data to a file on a TFTP or SFTP server.

| Parameter                         | Description                                                                                                                                                          |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>&lt;REMOTE-URL&gt;</code>   | Specifies the capture file on a remote TFTP or SFTP server. The URL syntax is:<br>{tftp://   sftp://<USER>@} {<IP>   <HOST>} [:<PORT>]<br>[;blocksize=<SIZE>]/<FILE> |
| <code>vrf &lt;VRF-NAME&gt;</code> | Specifies the name of a VRF. Default: default.                                                                                                                       |

## Example

Copying the capture data to a file on SFTP server 10.0.0.2:

```
switch# copy tshark-pcap sftp://root@10.0.0.2/file.pcap

root@10.0.0.2's password:
Connected to 10.0.0.2.
sftp> put packets.pcap file.pcap
Uploading packets.pcap to /root/file.pcap
packets.pcap 100% 156 219.8KB/s 00:00
Copied successfully.
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## destination cpu

```
destination cpu
no destination cpu
```

## Description

The command causes the mirror session to transmit mirrored packets to the switch CPU. This destination may be configured for multiple sessions, however only one such configured session may be active at a given time.

The diagnostic utility Tshark may be used to view and capture packets transmitted to the CPU through this route. Ctrl+C must be entered to terminate a Tshark capture session. More details can be found in the *Supportability Guide*.

The `no` form of this command will immediately stops mirroring traffic to the CPU, but will not remove any sources from the mirror configuration.

## Examples

Configuring a mirror session with CPU as the destination.

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination cpu
```

Removing the destination entirely.

```
switch(config-mirror-1)# no destination cpu
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context                          | Authority                                                                          |
|---------------|------------------------------------------|------------------------------------------------------------------------------------|
| All platforms | config-mirror- <i>&lt;SESSION-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## destination interface

```
destination interface {<INTERFACE-ID>|<LAG-NAME>}
no destination interface {<INTERFACE-ID>|<LAG-NAME>}
```

## Description

Configures the specified interface as the destination of the mirrored traffic.

The **no** form of this command immediately disables the mirroring session and removes the specified destination interface from the configuration.

| Parameter                   | Description                                          |
|-----------------------------|------------------------------------------------------|
| <i>&lt;INTERFACE-ID&gt;</i> | Specifies a interface. Format: member/slot/port.     |
| <i>&lt;LAG-NAME&gt;</i>     | Specifies a LAG (link aggregation group) identifier. |

## Usage

Supported mirror destinations: Layer 2 or Layer 3 Ethernet ports, LAGs, or CPU as a Mirror Destination. A port that is already a member of a LAG is not a valid mirror destination.

Configuring a different destination interface in an enabled mirroring session causes all mirrored traffic to use the new destination interface. This action might cause a temporary suspension of mirrored source traffic during the reconfiguration.

## Examples

Configuring a mirroring session and adding an interface as a destination:

```
switch(config)# mirror session 1
switch(config-mirror-1)# destination interface 1/1/1
```

Replacing the existing destination with different interface:

```
switch(config-mirror-1)# destination interface 1/1/12
```

Removing a destination:

```
switch(config-mirror-1)# no destination interface 1/1/12
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context            | Authority                                                                          |
|---------------|----------------------------|------------------------------------------------------------------------------------|
| All platforms | config-mirror-<SESSION-ID> | Administrators or local user group members with execution rights for this command. |

## destination tunnel

```
destination tunnel <TUNNEL-IPV4> source <SOURCE-IPv4-ADDR>
dscp <DSCP-VALUE> vrf <VRF-NAME>
```

```
no destination tunnel
```

## Description

Specifies the tunnel where all mirrored traffic for the session is transmitted. Only one tunnel destination is allowed per session.

You may configure multiple mirror sessions with the same source/destination IP address pair, however, only one of those sessions sharing the same source/destination IP address pair can be enabled at a given time.

ERSPAN is not supported leaving the switch by the OOB port. If VRF management is configured for an ERSPAN session, the session will be in "mirror\_err\_tunnel\_oob\_port\_not\_supported" operation status.

ERSPAN is not supported leaving the switch encapsulated within another tunnel (e.g. GRE IPv4). When the path to the destination IP address will leave via a tunnel, the session will be in "tunnel\_route\_resolution\_not\_populated" operation status.



---

The interface/LAG used to transmit ERSPAN packets should not be a source in the same mirror session.

---

The `no` form of this command will cease the use of the tunnel and disable the session.

| Parameter          | Description                                                                                                     |
|--------------------|-----------------------------------------------------------------------------------------------------------------|
| <TUNNEL-IPv4-ADDR> | Specifies the tunnel address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.               |
| <SOURCE-IPv4-ADDR> | Specifies the source address in IPv4 format (x.x.x.x), where x is a decimal number from 0 to 255.               |
| <DSCP-VALUE>       | Specifies the DSCP value to be carried within the DS field of ERSPAN packet header. Range: 0 to 63. Default: 0. |
| <VRF-NAME>         | Specifies a VRF name. Default: default.                                                                         |

## Examples

Creating a Mirror Session and adding tunnel destination, source, dscp, and VRF:

```
switch# config
switch(config)# mirror session 1
switch(config-mirror-1)# destination tunnel 1.1.1.1 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing tunnel destination:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 10 vrf
default
```

Replacing the existing destination with a different DSCP value:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf
default
```

Replacing the existing destination with a different VRF:

```
switch(config-mirror-1)# destination tunnel 11.12.13.14 source 2.2.2.2 dscp 2 vrf
newvrf
```

Removing the destination:

```
switch(config-mirror-1)# no destination tunnel
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context            | Authority                                                                          |
|----------------------|----------------------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-mirror-<SESSION-ID> | Administrators or local user group members with execution rights for this command. |

## diagnostic

diagnostic

```
diag utilities tshark [file]
diag utilities tshark [delete-file]
```

### Description

Captures packets from a mirror-to-cpu session, and save the most recent 32MB to pcap file which can then be copied and analyzed. When capturing a mirror-to-cpu session to a file, packets will not be dumped to the console.



The `diagnostic` command must be entered prior to the `diag utilities tshark` command.

Use the `delete-file` form of this command to delete the most recent capture file.

Since `file` and `delete-file` are optional, the behavior of the base command `diag utilities tshark` does **not** save anything to a file, and instead dumps the tshark session to the console until **CTRL + c** is entered.

| Parameter   | Description                                 |
|-------------|---------------------------------------------|
| file        | Saves captured packets to a temporary file. |
| delete-file | Deletes the most recent captured file.      |

### Example

Performing diagnostic:

```
switch# diagnostic

switch# diagnostic utilities tshark file
Inspecting traffic mirrored to the CPU until Ctrl-C is entered
^CEnding traffic inspection.
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## diag utilities tcpdump

```
diag utilities tcpdump [command <TEXT> | delete file <FILE-NAME> | list-files |
 vrf <VRF-NAME> | count <COUNT-NUM> | proto <PROTO-NUM> | host-ip <IP-ADDR> | source-ip
 <IP-ADDR> | destination-ip <IP-ADDR> | host-port <PORT> | source-port <PORT> |
 destination-port <PORT> | verbosity <LEVEL> | print <DATA> | ethernet-type <ETH-NUM>]
```

### Description

Captures traffic received or transmitted over a network.

| Parameter                | Description                                                                                                             |
|--------------------------|-------------------------------------------------------------------------------------------------------------------------|
| command <TEXT>           | Captures packets based on a specified tcpdump command string.                                                           |
| delete file <FILE-NAME>  | Deletes specified tcpdump list files.                                                                                   |
| list-files               | Lists all the tcpdump capture files saved on the device.                                                                |
| vrf <VRF-NAME>           | Captures packets on the specified VRF. If no VRF is named, the default is used.                                         |
| count <COUNT-NUM>        | Runs the tcpdump command until the specified number of packets are captured. Range: 1-2147483647.                       |
| proto <PROTO-NUM>        | Captures packets of a particular type based on IP protocol number. Range: 0-255.                                        |
| host-ip <IP-ADDR>        | Captures packets matching with the source or destination IP address.                                                    |
| source-ip <IP-ADDR>      | Captures packets from the specified IP address.                                                                         |
| destination-ip <IP-ADDR> | Captures packets sent to the specified IP address.                                                                      |
| host-port <PORT>         | Captures packets matching with the source or destination port.                                                          |
| source-port <PORT>       | Captures packets from the specified IP port.                                                                            |
| destination-port <PORT>  | Captures packets sent to the specified IP port.                                                                         |
| verbosity <LEVEL>        | Captures packets of the specified verbosity. Range: level1-level4. If no verbosity is specified, the default is level1. |
| print <DATA>             | Captures the data of each packet. The maximum is 262144 bytes                                                           |
| ethernet-type <ETH-NUM>  | Captures packets based on the particular ethernet type. Range: 0-65535.                                                 |

### Usage

- When using the `command` option, the only traffic captured will be packets that have been mirrored to the CPU.

- When using the `command` option, command line sanitization is performed to prevent options that may cause harm or security issues. The following options are blocked:
  - `-i/--interface`
  - `-Z`
  - `-B/--buffer-size`
  - `-C`
  - `-W`
  - `-Z/--relinquish-privileges`
- Non-word operators such as "&" or "|" are not allowed. Use boolean keywords such as "and," "or," and "not."
- When using `command -r` to read a file, do not provide any directory path characters. Use `list-files` command to get the list of file names currently saved on the device, and then use those file names.
- A total of four files can be saved at any given point on the device. Packet capture files are not saved after a failover or reboot, but can be saved to external storage using the `copy tcpdump-pcap` command.

## Examples

Inspecting traffic mirrored to the CPU via `tcpdump` and saving the output to `my_capture_file.pcap`:

```
switch# diag utilities tcpdump command -c 2 -x -w my_capture_file.pcap
Inspecting traffic mirrored to the CPU via tcpdump until Ctrl-C is entered.
2 packets captured
2 packets received by filter
0 packets dropped by kernel
Ending traffic capture.
```

Listing saved capture files:

```
switch# diag utilities tcpdump list-files
my_capture_file.pcap
```

Reading `my_capture_file.pcap`:

```
switch# diag utilities tcpdump command -r my_capture_file.pcap
reading from file /tmp/tcpdump/my_capture_file1.pcap, link-type EN10MB (Ethernet)
 1 11:59:34.047867 IP6 localhost.40318 > localhost.ntp: NTPv2, Reserved, length
12
 0x0000: 0000 0304 0006 0000 0000 0000 0000 86dd
 0x0010: 608a 7e47 0014 1140 0000 0000 0000 0000 `~G...@.....
 0x0020: 0000 0000 0000 0001 0000 0000 0000 0000
 0x0030: 0000 0000 0000 0001 9d7e 007b 0014 0027~.{... '
 0x0040: 1601 0001 0000 0000 0000 0000
 2 11:59:34.047915 IP6 localhost.ntp > localhost.40318: NTPv2, Reserved, length
12
 0x0000: 0000 0304 0006 0000 0000 0000 0000 86dd
 0x0010: 6b8d 23c5 0014 1140 0000 0000 0000 0000 k.#....@.....
 0x0020: 0000 0000 0000 0001 0000 0000 0000 0000
 0x0030: 0000 0000 0000 0001 007b 9d7e 0014 0027{~... '
 0x0040: d681 0001 c016 0000 0000 0000
```

Removing `my_capture_file.pcap`:

```
switch# diag utilities tcpdump delete-file my_capture_file.pcap
Successfully removed file
```

## Command History

| Release | Modification       |
|---------|--------------------|
| 10.08   | Command introduced |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | Manager (#)     | Administrators or local user group members with execution rights for this command. |

## disable (mirror session)

disable

### Description

Disables the mirroring session specified by the current command context.

### Usage

By default, mirroring sessions are disabled.

When a mirroring session is disabled, the `show mirror` command for that session ID shows an Admin Status of `disable` and an Operation Status of `disabled`.

### Example

Disabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# disable
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context                          | Authority                                                                          |
|---------------|------------------------------------------|------------------------------------------------------------------------------------|
| All platforms | config-mirror- <i>&lt;SESSION-ID&gt;</i> | Administrators or local user group members with execution rights for this command. |

## enable (mirror session)



enable

## Description

Enables the mirroring session for the current command context.

## Usage

By default, mirroring sessions are disabled.

When a mirroring session is enabled, the `show mirror` command for that session ID shows an `Admin Status of enable` and an `Operation Status of enabled`.

If sFlow is enabled on an interface and a mirroring session specifies the same interface as the source of received traffic (the source is configured with a direction of `rx` or `both`):

- The attempt to enable the mirroring session fails and an error is returned.



When adding, removing, or changing the configuration of a source interface in an enabled mirroring session, packets from other mirror sources using the same destination interface might be interrupted.

## Example

Configuring and enabling a mirroring session:

```
switch(config)# mirror session 3
switch(config-mirror-3)# source interface 1/1/2 rx
switch(config-mirror-3)# destination interface 1/1/3
switch(config-mirror-3)# comment Monitor router port ingress-only traffic
switch(config-mirror-3)# enable
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context                               | Authority                                                                          |
|---------------|-----------------------------------------------|------------------------------------------------------------------------------------|
| All platforms | <code>config-mirror-&lt;SESSION-ID&gt;</code> | Administrators or local user group members with execution rights for this command. |

## mirror session

```
mirror session <SESSION-ID>
no mirror session <SESSION-ID>
```

## Description

Creates a mirroring session configuration context or enters an existing mirroring session configuration context.

From this context, you can enter commands to configure and enable or disable the mirroring session.

The `no` form of this command removes an existing mirroring session from the configuration.

| Parameter    | Description                                     |
|--------------|-------------------------------------------------|
| <SESSION-ID> | Specifies the session identifier. Range: 1 to 4 |

## Examples

```
switch(config)# mirror session 1
switch(config-mirror-1)#

switch(config)# mirror session 3
switch(config-mirror-3)#

switch(config)# no mirror session 1
switch(config)#
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context | Authority                                                                          |
|---------------|-----------------|------------------------------------------------------------------------------------|
| All platforms | config          | Administrators or local user group members with execution rights for this command. |

## show mirror

```
show mirror [<SESSION-ID>] [vsx-peer]
```

## Description

Shows information about mirroring sessions. If <SESSION-ID> is not specified, then the command shows a summary of all configured mirroring sessions. If <SESSION-ID> is specified, then the command shows detailed information about the specified mirroring session.

| Parameter    | Description                                                                                                                                                                                                                      |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <SESSION-ID> | Specifies the session identifier. Range: 1 to 4                                                                                                                                                                                  |
| vsx-peer     | Shows the output from the VSX peer switch. If the switches do not have the VSX configuration or the ISL is down, the output from the VSX peer switch is not displayed. This parameter is available on switches that support VSX. |

## Usage

Admin Status indicates the configured status. Admin Status is one of the following values:

enable

The mirroring session is enabled.

disable

The mirroring session has been configured but not yet enabled, or has been disabled.

Operation Status indicates the status of the mirroring session. Operation Status is one of the following values:

`dest_doesnt_exist`

The configured destination interface is not found in the system. The mirroring session cannot be enabled.

`destination_shutdown`

The mirroring session is enabled, but the destination interface is shut down. No traffic can be monitored.

`disabled`

The mirroring session is disabled and is not in an error condition.

`enabled`

The mirroring session is enabled.

`external/driver_error`

An internal ASIC hardware error occurred.

`hit_active_sessions_capacity`

The mirroring session could not be enabled because the maximum number of supported mirroring sessions are already enabled.

`internal_error`

An invalid parameter was passed to the ASIC software layer.

`no_dest_configured`

The mirroring session does not have a destination interface configured.

`no_name_configured`

A software error occurred. The mirroring session does not have a session ID in its configuration.

`null_mirror`

A software error occurred. The session object reference is invalid.

`out_of_memory`

The system is out of memory, reboot recommended.

`tunnel_route_resolution_not_populated`

If the destination tunnel IP address is not reachable.

`unknown_error`

An unexpected error occurred.

## Examples

Showing summary information about all configured mirroring sessions:

```
switch# show mirror
ID Admin Status Operation Status
--- -
1 enable enabled
2 disable disabled
3 disable disabled
4 enable internal_error
```

Showing detailed information about a single mirroring session:

```
switch# show mirror 3
Mirror Session: 3
Admin Status: disable
Operation Status: disabled
Comment: Monitor router port ingress-only traffic
Source: interface 1/1/2 rx
Destination: interface 1/1/3
Output Packets: 0
Output Bytes: 0
switch#
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context             | Authority                                                                                                                                                              |
|---------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| All platforms | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## source interface

```
source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
no source interface {<PORT-NUM> | <LAG-NAME>} [<DIRECTION>]
```

## Description

Configures the specified interface (either an Ethernet port or a LAG) as a source of traffic to be mirrored. The `no` form of this command ceases mirroring traffic from the specified source interface and removes the source interface from the mirroring session configuration.

| Parameter   | Description                                                                                                                                         |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <PORT-NUM>  | Specifies a physical port on the switch. Use the format member/slot/port (for example, 1/3/1).                                                      |
| <LAG-NAME>  | Specifies the identifier for the LAG (link aggregation group).                                                                                      |
| <DIRECTION> | Selects the direction of traffic to be mirrored from this source interface. There is no default for this parameter. Valid values are the following: |
| both        | Mirror both transmitted and received packets.                                                                                                       |
| rx          | Mirror only received packets.                                                                                                                       |
| tx          | Mirror only transmitted packets.                                                                                                                    |

## Usage

There is a limit of source interfaces in each direction of a given mirror session:

| Switch | Source interface limit |
|--------|------------------------|
| 8320   | 128                    |
| 8325   | 128                    |
| 8360   | 64                     |

However, there is a practical limit to the amount of traffic that a mirror destination can transmit. For example, mirroring session with multiple 10G sources can overwhelm a single 10G destination. You can configure the same source interface in multiple mirroring sessions, if required.



---

When adding, removing, or changing the configuration of a source port in an enabled mirroring session, packets from other mirror sources using the same destination port might be interrupted.

---

## Examples

Configuring a mirrored traffic source interface:

```
switch(config-mirror-1)# source interface
LAG-NAME Enter a LAG name. For example, lag10
PORT-NUM Enter a port number
```

Creating a mirroring session and configuring a source interface to mirror both transmitted and received packets:

```
switch(config)# mirror session 1
switch(config-mirror-1)# source interface 1/1/1 both
```

Creating a second mirroring session and configuring two source interfaces. One port mirroring only transmitted packets and the other mirroring both transmitted and received packets:

```
switch(config)# mirror session 2
switch(config-mirror-2)# source interface 1/1/3 tx
switch(config-mirror-2)# source interface 1/2/1 both
```

Removing the first source interface:

```
switch(config-mirror-2)# no source interface 1/2/3
```

Configuring a source interface to mirror received packets only:

```
switch(config-mirror-3)# source interface 1/1/2 rx
```

Configuring a source interface to mirror both transmitted and received packets:

```
switch(config-mirror-1)# source interface 1/1/1 both
```

Configuring a LAG as source interface to mirror both transmitted and received packets:

```
switch(config-mirror-4)# source interface lag1 both
```

Stopping the mirroring of received packets from a configured source interface:

```
switch(config-mirror-4)# no source interface lag1 rx
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms     | Command context            | Authority                                                                          |
|---------------|----------------------------|------------------------------------------------------------------------------------|
| All platforms | config-mirror-<SESSION-ID> | Administrators or local user group members with execution rights for this command. |

## source vlan



Applies only to 8360.

### Syntax

```
source vlan <VLAN-NUM> {rx | tx | both}
no source vlan <VLAN-NUM> [rx | tx | both]
```

### Description

Adds or removes VLAN as a source of traffic to be mirrored. More than one source VLAN can be configured in a mirror session. Each VLAN may specify its own direction.

The `no` version of the command ceases mirroring traffic from the specified source VLAN and removes the source from the mirror configuration.

There is a limit of 1024 source VLANs in each direction of a given mirror session. The same VLAN can be configured as a mirror source for multiple sessions.

### Command context

config

### Parameters

<VLAN-NUM>

Configured VLAN number.

rx

Mirror only received traffic.

tx

Mirror only transmitted traffic.

both

Mirror both received and transmitted traffic.

### Authority

Administrators or local user group members with execution rights for this command.

### Example

Create a mirror session and add VLAN 10 as a source of traffic in both directions on that port.

```
switch(config)# mirror session 1
switch(config-mirror-1)# source vlan 10 both
```

Create a second mirror session and add VLAN 10 as a transmit sources of traffic and VLAN 20 in both receive and transmit directions.

```
switch(config)# mirror session 2
switch(config-mirror-2)# source vlan 10 tx
switch(config-mirror-2)# source vlan 20 both
```

Reconfigure the source in session 2 to be receive only by respecifying the source interface configuration.

```
switch(config-mirror-2)# source vlan 10 rx
```

From the second session, remove the first source interface entirely and remove the transmit direction from the other so that mirroring only occurs in the receive direction.

```
switch(config-mirror-2)# no source vlan 10
switch(config-mirror-2)# no source vlan 20 tx
```

Message received when trying to add more than 1024 mirror source VLANs

```
switch(config-mirror-2)# source vlan 2000 rx
The maximum number of source VLANs per mirror session is 1024 in each direction
```

**Configuring SNMP:** Refer to the *SNMP/MIB Guide* for information on how to add SNMP so a device can be monitored from a network management system (NMS).

**Configuring an SNMP trap receiver:** Refer to the *SNMP/MIB Guide* and specific information about the `show snmp trap` command to enable SNMP traps.

### PoE commands

All PoE configuration commands except `threshold` configuration and `always-on poe` configuration are entered at the `config-if` context. The PoE `threshold` command is used at the system level whereas the `always-on poe` and `power-over-ethernet quick-poe` commands are set at the slot level. These commands can only be configured in the global configuration context.

### lldp dot3 poe

```
lldp dot3 poe
no lldp dot3 poe
```

#### Description

Enables 802.3 TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. LLDP dot3 TLV is by default enabled for PoE.

The `no` form of this command disables 802.3 TLV list in LLDP.

#### Examples

Enabling 802.3 TLV list in LLDP:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp dot3 poe
```

Disabling 802.3 TLV list in LLDP:

```
switch(config-if)# no lldp dot3 poe
```

#### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

#### Command Information



| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## lldp med poe

```
lldp med poe [priority-override]
no lldp med poe [priority-override]
```

### Description

Enables MED TLV list in LLDP to advertise for Power over Ethernet Data Link Layer Classification. Also enables the lldp-MED TLV priority to override user configured port priority for Power over Ethernet. When both dot3 and MED are enabled, dot 3 will take precedence. MED TLV is by default enabled for PoE. Priority over-ride is by default disabled.

The `no` form of this command disables MED TLV list in LLDP.

| Parameter           | Description                           |
|---------------------|---------------------------------------|
| [priority-override] | System defined name of the interface. |

### Examples

Enabling and disabling LLDP MED PoE:

```
switch(config)# interface 1/1/1
switch(config-if)# lldp med poe
switch(config-if)# no lldp med poe
```

Enabling and disabling LLDP MED PoE priority override:

```
switch(config-if)# lldp med poe priority-override
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet

```
power-over-ethernet
no power-over-ethernet
```

### Description

Enables per-interface power distribution. Per-port power is enabled by default with priority low. PoE cannot be disabled for individual ports when Quick PoE is enabled for the entire switch or line module. The `no` form of this command disables per-interface power distribution.

## Examples

Enabling per-interface power distribution:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet
```

Disabling per-interface power distribution:

```
switch(config-if)# no power-over-ethernet
```

Showing Quick PoE enabled:

```
switch(config-if)# power-over-ethernet quick-poe 1/1
switch(config-if)# interface 1/1/1
switch(config-if)# no power-over-ethernet
Interface PoE cannot be disabled when Quick PoE is enabled.
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet allocate-by

```
power-over-ethernet allocate-by {usage | class}
no power-over-ethernet allocate-by {usage | class}
```

## Description

Configures the power allocation method. Power allocation method is initially based on usage. PSE Allocated power value will change to LLDP negotiated power if and when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocated Power Value will be the actual instantaneous power draw and reserve power based on actual consumption. In allocate-by class, power allocation is based on PD requested class and PSE allocated power value will be the LLDP negotiated power when LLDP exchange takes place between PSE and PD. When there is no LLDP negotiation, PSE Allocate Power will be based on PD class. Reserve power is based on PD Class. By default, power allocation is by usage.

The `no` form of this command resets the action to default.

## Examples

Configuring the power allocation method:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet allocate-by usage
switch(config-if)# power-over-ethernet allocate-by class
```

Resetting power allocation method:

```
switch(config-if)# no power-over-ethernet allocate-by class
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet always-on

```
power-over-ethernet always-on <MODULE-ID>
no power-over-ethernet always-on <MODULE-ID>
```

## Description

Always-on PoE is a feature that provides the ability to the switch to continue to provide power across a soft reboot. It is applicable only to the interfaces which were connected and delivering before the soft reboot. Also, power will not be delivered if power to the switch is interrupted. This command enables or disables the always-on PoE feature at the switch or the slot level. By default, always-on PoE is enabled at the switch or the slot level.

The `no` form of this command disables power distribution on soft reboot.

| Parameter   | Description                                         |
|-------------|-----------------------------------------------------|
| <MODULE-ID> | Module number to apply always-on PoE configuration. |

## Examples

Enabling per-interface power distribution:

```
switch(config)# power-over-ethernet always-on 1/1
```

Disabling per-interface power distribution:

```
switch(config)# no power-over-ethernet always-on 1/1
```

## Command History

| Release | Modification |
|---------|--------------|
|---------|--------------|

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config          | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet assigned-class

```
power-over-ethernet assigned-class {3 | 4 | 6}
no power-over-ethernet assigned-class
```

### Description

Limit PoE power based on the assigned class. When an user assigns a maximum class to an interface, the PSE will limit the maximum power delivered to the PD up to a total power draw not exceeding the PSE assigned-class power. Power demotion occurs when a PD requested class is higher than the PSE assigned class, permitting the PD to receive power and operate in a reduced power mode. PoE ports cannot set an assigned class when Quick PoE is enabled on the subsystem. The default assigned class is 4 for 2-pair capable PSE and 6 for 4-pair capable PSE.

The `no` form of this command resets the action to default.

### Examples

Setting PoE assigned class:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet assigned-class 4
```

Resetting PoE assigned class to default:

```
switch(config-if)# no power-over-ethernet assigned-class 4
```

Showing Quick PoE enabled:

```
switch(config)# power-over-ethernet quick-poe 1/1
switch(config)# interface 1/1/1
switch(config)# power-over-ethernet assigned-class 4
Interface assigned class cannot be configured when Quick PoE is enabled.
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet power-pairs

```
power-over-ethernet power-pairs {alt-a | alt-a-and-alt-b}
no power-over-ethernet power-pairs {alt-a | alt-a-and-alt-b}
```

### Description

Configures the four-pair capable switch to operate in a mode, that restricts the power delivery for class 0 to class 4 single signature devices to operate only on ALT-A power pair.



When configured, a warning message is displayed. User must accept the warning by entering `y` to enable the mode.

The `no` form of this command resets the power pairs to default PoE pairs.

| Parameter                    | Description                                                                                           |
|------------------------------|-------------------------------------------------------------------------------------------------------|
| <code>alt-a</code>           | Delivers power only on the ALT-A pair.                                                                |
| <code>alt-a-and-alt-b</code> | Delivers power on the ALT-A and ALT-B pairs. This is the default configuration on all PoE interfaces. |

### Usage

IEEE 802.3bt devices such as four-pair (class 5 and higher) and dual signature powered devices require power on both pairs. However, there is no such restriction on IEEE 802.3af (class 0 to class 3) and IEEE 802.3at (class 4) powered devices not to draw power on both pairs if the overall consumption does not violate the power class limit. For such powered devices, a `power-pairs` configuration is provided to configure the 4-pair capable switch to restrict power on only one power pair.

### Examples

Configuring PoE power pairs:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet power-pairs alt-a
```

This setting configures the interface to deliver power only on the ALT-A cable pair when a Class 0-4 device is connected. Devices that require power on all pairs may not operate correctly.

```
Continue (y/n)? y
```

Resetting the PoE power pair to default:

```
switch(config-if)# no power-over-ethernet power-pairs alt-a
```

This setting configures the interface to deliver power on the ALT-A and ALT-B cable pairs. This is the default and most devices work properly with this setting, however some older Class 0-4 devices may not operate correctly.

Continue (y/n)? y

## Command History

| Release | Modification       |
|---------|--------------------|
| 10.09   | Command Introduced |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet pre-std-detect

```
power-over-ethernet pre-std-detect
no power-over-ethernet pre-std-detect
```

## Description

Before IEEE 802.3 released the first Power over Ethernet standard (802.3af), vendors had shipped PoE capable switches and PD's. As we are backward compatible Aruba will support both IEEE standard and pre-standard 802.3af Power over Ethernet PD's concurrently. This CLI allows the user to enable or disable pre-802.3af-standard device detection and powering on the specific port. When pre-std-detect is enabled, power will be delivered on PairA only. Default is disabled.

The `no` form of this command resets the action to default.

## Examples

Enabling standard device detection:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet pre-std-detect
```

Disabling standard device detection:

```
switch(config-if)# no power-over-ethernet pre-std-detect
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet priority

```
power-over-ethernet priority {critical | high | low}
no power-over-ethernet priority {critical | high | low}
```

### Description

Sets PoE priority for an interface Specifying critical, high, or low indicates the priority of the interface in the event of power over-subscription. Within the same priority level, higher power-priority line-module ports have higher precedence. With same PoE priority and same line-module priority, lower numbered line-module ports have higher precedence. Per-interface PoE priority is low by default.

The `no` form of this command resets the priority to default PoE priority "low".

### Examples

Configuring PoE priority:

```
switch(config)# interface 1/1/1
switch(config-if)# power-over-ethernet priority critical
switch(config-if)# power-over-ethernet priority high
```

Resetting the PoE priority to default:

```
switch(config-if)# no power-over-ethernet priority high
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet quick-poe

```
power-over-ethernet quick-poe <MODULE-ID>
no power-over-ethernet
```

## Description

Quick PoE is a feature that provides the ability for the switch to provide power to the connected powered device as soon as switch goes through cold reboot. When quick PoE is enabled on the subsystem PoE port disablement and PD demotion is not allowed. also quick PoE enablement is not allowed if any of the port is disabled on the subsystem. User should not over-subscribe the PoE power when quick PoE is enabled. Quick PoE saved configuration will work irrespective of the configuration change at reboot.

Enables quick PoE feature on the switch or the subsystem level. By default, quick-PoE is disabled for the subsystem.

The `no` form of this command disables quick PoE.

| Parameter                      | Description                                           |
|--------------------------------|-------------------------------------------------------|
| <code>&lt;MODULE-ID&gt;</code> | Specifies module number for quick PoE configuration . |

## Examples

Enabling and disabling quick PoE:

```
switch(config)# power-over-ethernet quick-poe 1/2
switch(config)# no power-over-ethernet quick-poe 1/2
```

```
switch(config-if)# power-over-ethernet quick-poe 1/1
PoE must be enabled on all interfaces before enabling Quick PoE
```

```
switch(config-if)# power-over-ethernet quick-poe 1/3
All interfaces must use the default assigned class before enabling Quick PoE
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet threshold

```
power-over-ethernet threshold <PERCENTAGE>
no power-over-ethernet threshold <PERCENTAGE>
```

## Description

Sets the threshold at which the system will send an excess power consumption notification trap. Default value is 80 percentage.



The `no` form of this command resets the action to default.

| Parameter    | Description                                          |
|--------------|------------------------------------------------------|
| <PERCENTAGE> | Excess power consumption trap threshold. Range 1-99. |

## Examples

Setting the power-over-ethernet threshold:

```
switch(config)# power-over-ethernet threshold 75
```

Resetting the power-over-ethernet threshold to default:

```
switch(config-if)# no power-over-ethernet threshold 75
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config          | Administrators or local user group members with execution rights for this command. |

## power-over-ethernet trap

```
power-over-ethernet trap
no power-over-ethernet trap
```

### Description

This command enables/disables the SNMP trap generation for PoE related events at system level. PoE trap generation is enabled by default.

The `no` form of this command resets the priority to default PoE priority "low".

### Examples

Enabling SNMP trap generation for PoE:

```
switch(config)# power-over-ethernet trap
```

Disabling SNMP trap generation for PoE:

```
switch(config-if)# no power-over-ethernet trap
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context | Authority                                                                          |
|-----------|-----------------|------------------------------------------------------------------------------------|
|           | config-if       | Administrators or local user group members with execution rights for this command. |

## show lldp local

show lldp local-device [<INTERFACE-ID>]

### Description

Displays information advertised by the switch if the LLDP feature is enabled by user.

| Parameter      | Description                                      |
|----------------|--------------------------------------------------|
| <INTERFACE-ID> | Specifies an interface. Format: member/slot/port |

### Examples

Showing LLDP local device:

```
switch# show lldp local-device 1/1/10
Local Port Data
=====

Port-ID : 1/1/10
Port-Desc : "1/1/10"
Port VLAN ID : 0

PoE Plus Information

PoE Device Type : Type 2 PSE
Power Source : Primary
Power Priority : low
PSE Allocated Power: 25.0 W
PD Requested Power : 25.0 W
```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show lldp neighbor

show lldp neighbor [<INTERFACE-ID>]

### Description

Displays detailed information about a particular neighbor connected to a particular interface.

| Parameter      | Description                                      |
|----------------|--------------------------------------------------|
| <INTERFACE-ID> | Specifies an interface. Format: member/slot/port |

### Examples

Showing LLDP neighbor information when there is only one neighbor:

```
switch# show lldp neighbor-info 1/1/10

Port : 1/1/10
Neighbor Entries : 1
Neighbor Entries Deleted : 0
Neighbor Entries Dropped : 0
Neighbor Entries Aged-Out : 0
Neighbor Chassis-Name : 84:d4:7e:ce:5d:68
Neighbor Chassis-Description : ArubaOS (MODEL: 325), Version Aruba IAP
Neighbor Chassis-ID : 84:d4:7e:ce:5d:68
Neighbor Management-Address : 169.254.41.250
Chassis Capabilities Available : Bridge, WLAN
Chassis Capabilities Enabled :
Neighbor Port-ID : 84:d4:7e:ce:5d:68
Neighbor Port-Desc : eth0
TTL : 120
Neighbor Port VLAN ID :
Neighbor PoEplus information : DOT3
Neighbor Device Type : TYPE2 PD
Neighbor Power Priority : Unkown
Neighbor Power Source : Primary
Neighbor Power Requested : 25.0 W
Neighbor Power Allocated : 0.0 W
Neighbor Power Supported : No
Neighbor Power Enabled : No
Neighbor Power Class : 5
Neighbor Power Paircontrol : No
Neighbor Power Pairs : SIGNAL
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

## show power-over-ethernet

### Description

Displays the status information of the full system.

| Parameter | Description                                              |
|-----------|----------------------------------------------------------|
| brief     | Display the brief status of all ports or the given port. |

### Examples

Showing sample output for power-over-ethernet brief per-port:

```
switch# show power-over-ethernet 1/1/1 brief

Status and Configuration Information for port 1/1/1

Member 1Power Status
Available: 370 W Reserved: 55.60 W Remaining: 314.40 W
Always-on PoE Enabled: 1/1
PoE Pwr Power Pre-std Alloc PSE Pwr PD Pwr PoE Port PD Cls Type
Port En Priority Detect Act Rsrvd Draw Status Sign --- ----

1/1/1 Yes Low Off Class 0.0 W 0.0 W Denied None 4 2
```

Showing sample output for power-over-ethernet for a missing line card:

```
switch# show power-over-ethernet 1/3 brief

Module 1/3 is not physically present.
```

Showing sample output for power-over-ethernet port when physical interface is not present:

```
switch# show power-over-ethernet 2/1/1

Interface 2/1/1 is not present.
```

### Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

### Command Information

| Platforms | Command context             | Authority                                                                                                                                                              |
|-----------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|           | Operator (>) or Manager (#) | Operators or Administrators or local user group members with execution rights for this command. Operators can execute this command from the operator context (>) only. |

Ports default to an unsplit state. When a port is 'split', the split interfaces become active and can be configured independently. For example, when a 40G QSFP+ port is split four ways, each split interface behaves like a separate 10G SFP+ port. The split interfaces have the same name as the base port with an added suffix to represent their lane of the breakout cable or optical channel on SR4 optics. Splitting an interface removes most of the port's configuration settings and makes it inactive. The port will no longer appear in many show interface commands and most configuration commands are not allowed; the split interface name must be used.

The same thing happens in reverse when an interface is unsplit. However, note that the 'split' and 'no split' commands are always performed in the unsplit port's context.

## Limitations with breakout cable support

- The JL720A Aruba 8360-48XT4C models (ordered SKU #s JL706A/JL707A) do not support split ports.

## Breakout cable support commands

### split

```
split [confirm]
no split [confirm]
```

### Description

Splits a port into multiple interfaces. Only ports capable of supporting breakout cables or SR4/eSR4 optics can be split.

| Parameter | Description                                   |
|-----------|-----------------------------------------------|
| confirm   | Specifies the confirmation of port splitting. |

### Usage

The splittable ports for all models are shown in the table below:

| Part Number (PN)                                                                                                    | Description                                                                                               | Port info                                                  |
|---------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|------------------------------------------------------------|
| <b>Aruba 8320 Series</b> <ul style="list-style-type: none"><li>▪ JL479A</li><li>▪ JL579A</li><li>▪ JL581A</li></ul> | Aruba 8320 48 10/6 40 X472 5 2 Bdl<br>Aruba 8320 32 40G X472 5 2 Bdl<br>Aruba 8320 48 T/6 40 X472 5 2 Bdl | 49-54 (40G)<br>5-28 (40G - center 24 ports)<br>49-54 (40G) |

| Part Number (PN)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Port info                                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Aruba 8325 48Y8C models</b><br>JL635A (base system) <ul style="list-style-type: none"> <li>■ JL624A - Port-to-Power model (FB)</li> <li>■ JL625A - Power-to-Port model (BF)</li> </ul> <b>Aruba 8325 32C models</b><br>JL636A (base system) <ul style="list-style-type: none"> <li>■ JL626A - Port-to-Power model (FB)</li> <li>■ JL627A - Power-to-Port model (BF)</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Displayed by <code>show system</code><br>Aruba 8325-48Y8C FB 6 F 2 PS Bdl<br>Aruba 8325-48Y8C BF 6 F 2 PS Bdl<br><br>Displayed by <code>show system</code><br>Aruba 8325-32C FB 6 F 2 PS Bdl<br>Aruba 8325-32C BF 6 F 2 PS Bdl                                                                                                                                                                                                                                                                                                                                                                              | 49-56 (40G or 100G)<br>49-56 (40G or 100G)<br><br>1-32 (40G or 100G)<br>1-32 (40G or 100G)                                                                                                                                       |
| <b>Aruba 8360 32Y4C models</b><br>JL717A (base system) <ul style="list-style-type: none"> <li>■ JL700A Port-to-Power model</li> <li>■ JL701A Power-to-Port model</li> </ul> <b>Aruba 8360 16Y2C models</b><br>JL718A (base system) <ul style="list-style-type: none"> <li>■ JL702A Port-to-Power model</li> <li>■ JL703A Power-to-Port model</li> </ul> <b>Aruba 8360 48XT4C models</b><br>JL720A (base system) <ul style="list-style-type: none"> <li>■ JL706A Port-to-Power model</li> <li>■ JL707A Power-to-Port model</li> </ul> <b>Aruba 8360-12C models</b><br>JL721A (base system) <ul style="list-style-type: none"> <li>■ JL708A Port-to-Power model</li> <li>■ JL709A Power-to-Port model</li> </ul> <b>Aruba 8360 24XF2C models</b><br>JL722A (base system) <ul style="list-style-type: none"> <li>■ JL710A Port-to-Power model</li> <li>■ JL711A Power-to-Port model</li> </ul> | Displayed by <code>show system</code><br>Aruba 8360-32Y4C Prt2Pwr3F2PS Bdl<br>Aruba 8360-32Y4C Pwr2Prt3F2PS Bdl<br><br>Displayed by <code>show system</code><br>Aruba 8360-16Y2C Prt2Pwr3F2PS Bdl<br>Aruba 8360-16Y2C Pwr2Prt3F2PS Bdl<br><br>Displayed by <code>show system</code><br>Aruba 8360-48XT4C Prt2Pwr3F2PS Bdl<br>Aruba 8360-48XT4C Pwr2Prt3F2PS Bdl<br><br>Displayed by <code>show system</code><br>Aruba 8360-12C Pwr2Prt3F2PS Bdl<br>Aruba 8360-12C Prt2Pwr3F2PS Bdl<br><br>Displayed by <code>show system</code><br>Aruba 8360-24XF2C Prt2Pwr3F2PS Bdl<br>Aruba 8360-24XF2C Pwr2Pwr3F2PS Bdl | 33-36 (40G or 100G)<br>33-36 (40G or 100G)<br><br>17-18 (40G or 100G)<br>17-18 (40G or 100G)<br><br>NO SUPPORT for Split ports<br><br>1-12 (40G or 100G)<br>1-12 (40G or 100G)<br><br>25-26 (40G or 100G)<br>25-26 (40G or 100G) |



For releases 10.05 and 10.06, the 8320, 8325, and 8360 Switch Series require a save and reboot to enable the split port. From release 10.07 onwards these switches no longer require a save and reboot.

## Examples

Before splitting an interface:

```
8325(config)# show interface 1/1/56 brief
```

```

--
Port Native Mode Type Enabled Status Reason Speed
Desc VLAN
(Mb/s)
```

```

--
1/1/56 -- routed QSFP+DA1 no down Administratively down -- --

```

After splitting:

```

8325(config)# interface 1/1/56
8325(config-if)# split
This command will disable the specified port, clear its configuration,
and split it into multiple interfaces.

Continue (y/n)? y

8325(config-if)# show interface 1/1/56,1/1/56:1-1/1/56:4 brief

--
Port Native Mode Type Enabled Status Reason Speed
Desc
 VLAN

--
1/1/56 -- routed QSFP+DA1 no down Interface split -- --
1/1/56:1 -- routed QSFP+DA1 yes up -- 10000 --
1/1/56:2 -- routed QSFP+DA1 yes up -- 10000 --
1/1/56:3 -- routed QSFP+DA1 yes up -- 10000 --
1/1/56:4 -- routed QSFP+DA1 yes up -- 10000 --

```

Unsplitting a port on a switch that does not require a reboot:

```

switch(config)# interface 1/1/1
switch(config-if)# no split
This command will disable the split interfaces for this port and clear
their configuration.

Continue (y/n)? y

```

## Command History

| Release          | Modification |
|------------------|--------------|
| 10.07 or earlier | --           |

## Command Information

| Platforms            | Command context | Authority                                                                          |
|----------------------|-----------------|------------------------------------------------------------------------------------|
| 8320<br>8325<br>8360 | config-if       | Administrators or local user group members with execution rights for this command. |



You can manage and monitor the AOS-CX switch through Aruba AirWave. The following benefits and functions include:

- Configuration (partial configuration)
- Device topology
- Immediate and historical trend reports
- Monitoring of the device and user connected to the network.
- Network discovery
- Syslogs and trap receiver

For information about which versions of Aruba AirWave support AOS-CX, see the *AOS-CX Release Notes*.

## SNMP support and AirWave

For AirWave to discover and monitor the switch, you must:

- Enable the SNMP services on the switch.
- Configure the SNMP agent to use the SNMP version supported by the management station.

### SNMP on the switch

The switch provides SNMP services through the management channel and the data interfaces. Functionality, such as device discovery from NMS, syslog and trap forwarding, can be any channel configured by you.

Although the SNMP server can be enabled on both VRFs (`mgmt` and `default`), only one instance of SNMP can be running. The highest priority is on the `default` VRF.

For example, assume that SNMP is first enabled on the `mgmt` VRF (`snmp-server vrf mgmt`). Then, SNMP is enabled on the `default` VRF (`snmp-server vrf default`) without disabling SNMP on the `mgmt` (using an equivalent `no` form of the command). The `show running-config` command displays both `snmp-server vrf` commands; however, the SNMP instance is running only on the `default` VRF (highest priority).

```
switch# config
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
switch(config)# show running-config
Current configuration:
!
!Version AOS-CX Virtual.10.01.
led locator on
!
!
!
snmp-server vrf default
snmp-server vrf mgmt
```

!  
...

## Supported features with AirWave and the AOS-CX switch

AirWave supports the following features with the AOS-CX switch:

|                          |                                                                                                                                                               |
|--------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Device management        | Device discovery using SNMPv2C and SNMPv3                                                                                                                     |
|                          | Device dashboards                                                                                                                                             |
| Monitoring management    | Device health attributes (device status/reachability)                                                                                                         |
|                          | Interface and VLAN management                                                                                                                                 |
|                          | Initiates an SSH connection from Aruba AirWave to AOS-CX so that the device outputs from the AOS-CX CLI can be displayed in the Aruba AirWave user interface. |
|                          | Firmware versions                                                                                                                                             |
|                          | Displays neighbor devices connected to AOS-CX switches                                                                                                        |
|                          | Device topology                                                                                                                                               |
| Configuration management | Partial configuration                                                                                                                                         |
| Alarm management         | Alarm triggers (device and interface up/down, new device discoveries, custom event triggers)                                                                  |
|                          | Syslogs and traps                                                                                                                                             |
| Report management        | Device inventory, interface utilization, and device reachability reports                                                                                      |
|                          | Summary report of device model, firmware, and boot loader version                                                                                             |

## Configuring the AOS-CX switch to be monitored by AirWave

### Prerequisites

Aruba AirWave is active on the network.

### Procedure

1. Enable SNMP on the switch by entering the `snmp-server vrf mgmt` command.

```
switch(config)# snmp-server vrf mgmt
switch(config)# snmp-server vrf default
```

2. Configure the SNMPv2C community to public by entering the `snmp-server community public` command. In this instance, `public` is a read-only community string.

```
switch(config)# snmp-server community public
```

3. The community-string is used by SNMPv1 and SNMPv2C for unencrypted authentication. SNMPv3 lets you encrypt the authentication mechanism. To enable SNMPv3, enter the `snmpv3 user` and `snmpv3 context` commands.

```
switch(config)# snmpv3 user Admin auth sha auth-pass ciphertext
AQBapZHf2d20GYr/xcGUzYzm0zjNf/4VKHtSqBNImqtfYbJYCgAAALkGFJVcSp3nZ3o=
priv des priv-pass ciphertext
AQBapb0H2poBQKXPoVsC9L9qzZyfJQnzR7hmTr7LGsOsI7K3CgAAAKP98Rq2jfTrFwQ=
switch(config)# snmpv3 context Admin
```

For discovering devices in AirWave through the SNMPv3 community, the SNMPv3 context name is not mandatory. Devices can still be discovered in Aruba AirWave without the SNMPv3 context name.

4. Enter the `logging` command for enabling syslog forwarding to a remote syslog server, such as AirWave:

```
switch(config)# logging 10.0.10.2 severity debug
```

5. SNMP traps enable an agent to notify the management station of significant events by way of an unsolicited SNMP message. Enable SNMP traps by entering the `snmp-server host` command:

```
switch(config)# snmp-server host 10.10.10.10 trap version v2c vrf default
```

SNMP traps cannot be forwarded from AOS-CX 10.00 switches that have the VRF configured as `mgmt`. Later versions of AOS-CX support SNMP trap forwarding even when the VRF is configured as `default` or `mgmt`.

6. For information on how to add a device for monitoring in the Aruba AirWave user interface, see the documentation for Aruba AirWave.

### Accessing Aruba Support

|                         |                                                                                                                                                                                      |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Aruba Support Services  | <a href="https://www.arubanetworks.com/support-services/">https://www.arubanetworks.com/support-services/</a>                                                                        |
| Aruba Support Portal    | <a href="https://asp.arubanetworks.com/">https://asp.arubanetworks.com/</a>                                                                                                          |
| North America telephone | 1-800-943-4526 (US & Canada Toll-Free Number)<br>+1-408-754-1200 (Primary - Toll Number)<br>+1-650-385-6582 (Backup - Toll Number - Use only when all other numbers are not working) |
| International telephone | <a href="https://www.arubanetworks.com/support-services/contact-support/">https://www.arubanetworks.com/support-services/contact-support/</a>                                        |

Be sure to collect the following information before contacting Support:

- Technical support registration number (if applicable)
- Product name, model or version, and serial number
- Operating system name and version
- Firmware version
- Error messages
- Product-specific reports and logs
- Add-on products or components
- Third-party products or components

#### Other useful sites

Other websites that can be used to find information:

|                                                      |                                                                                                                                                                                         |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Airheads social forums and Knowledge Base            | <a href="https://community.arubanetworks.com/">https://community.arubanetworks.com/</a>                                                                                                 |
| AOS-CX Switch Software Documentation Portal          | <a href="https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm">https://www.arubanetworks.com/techdocs/AOS-CX/help_portal/Content/home.htm</a>                     |
| Aruba Hardware Documentation and Translations Portal | <a href="https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm">https://www.arubanetworks.com/techdocs/hardware/DocumentationPortal/Content/home.htm</a> |

|                         |                                                                                                                                       |
|-------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| Aruba software          | <a href="https://asp.arubanetworks.com/downloads">https://asp.arubanetworks.com/downloads</a>                                         |
| Software licensing      | <a href="https://lms.arubanetworks.com/">https://lms.arubanetworks.com/</a>                                                           |
| End-of-Life information | <a href="https://www.arubanetworks.com/support-services/end-of-life/">https://www.arubanetworks.com/support-services/end-of-life/</a> |
| Aruba Developer Hub     | <a href="https://developer.arubanetworks.com/">https://developer.arubanetworks.com/</a>                                               |

## Accessing Updates

You can access updates from the Aruba Support Portal or the HPE My Networking Website.

### Aruba Support Portal

<https://asp.arubanetworks.com/downloads>

If you are unable to find your product in the Aruba Support Portal, you may need to search My Networking, where older networking products can be found:

### My Networking

<https://www.hpe.com/networking/support>

To view and update your entitlements, and to link your contracts and warranties with your profile, go to the Hewlett Packard Enterprise Support Center **More Information on Access to Support Materials** page:

<https://support.hpe.com/portal/site/hpsc/aae/home/>

Access to some updates might require product entitlement when accessed through the Hewlett Packard Enterprise Support Center. You must have an HP Passport set up with relevant entitlements.

Some software products provide a mechanism for accessing software updates through the product interface. Review your product documentation to identify the recommended software update method.

To subscribe to eNewsletters and alerts:

<https://asp.arubanetworks.com/notifications/subscriptions> (requires an active Aruba Support Portal (ASP) account to manage subscriptions). Security notices are viewable without an ASP account.

## Warranty Information

To view warranty information for your product, go to <https://www.arubanetworks.com/support-services/product-warranties/>.

## Regulatory Information

To view the regulatory information for your product, view the *Safety and Compliance Information for Server, Storage, Power, Networking, and Rack Products*, available at <https://www.hpe.com/support/Safety-Compliance-EnterpriseProducts>

### Additional regulatory information

Aruba is committed to providing our customers with information about the chemical substances in our products as needed to comply with legal requirements, environmental data (company programs,

product recycling, energy efficiency), and safety information and compliance data, (RoHS and WEEE). For more information, see <https://www.arubanetworks.com/company/about-us/environmental-citizenship/>.

## Documentation Feedback

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